

이미지처리팀

Self-Supervised Transformers for Unsupervised Object Discovery using Normalized Cut

CVPR22 Paper

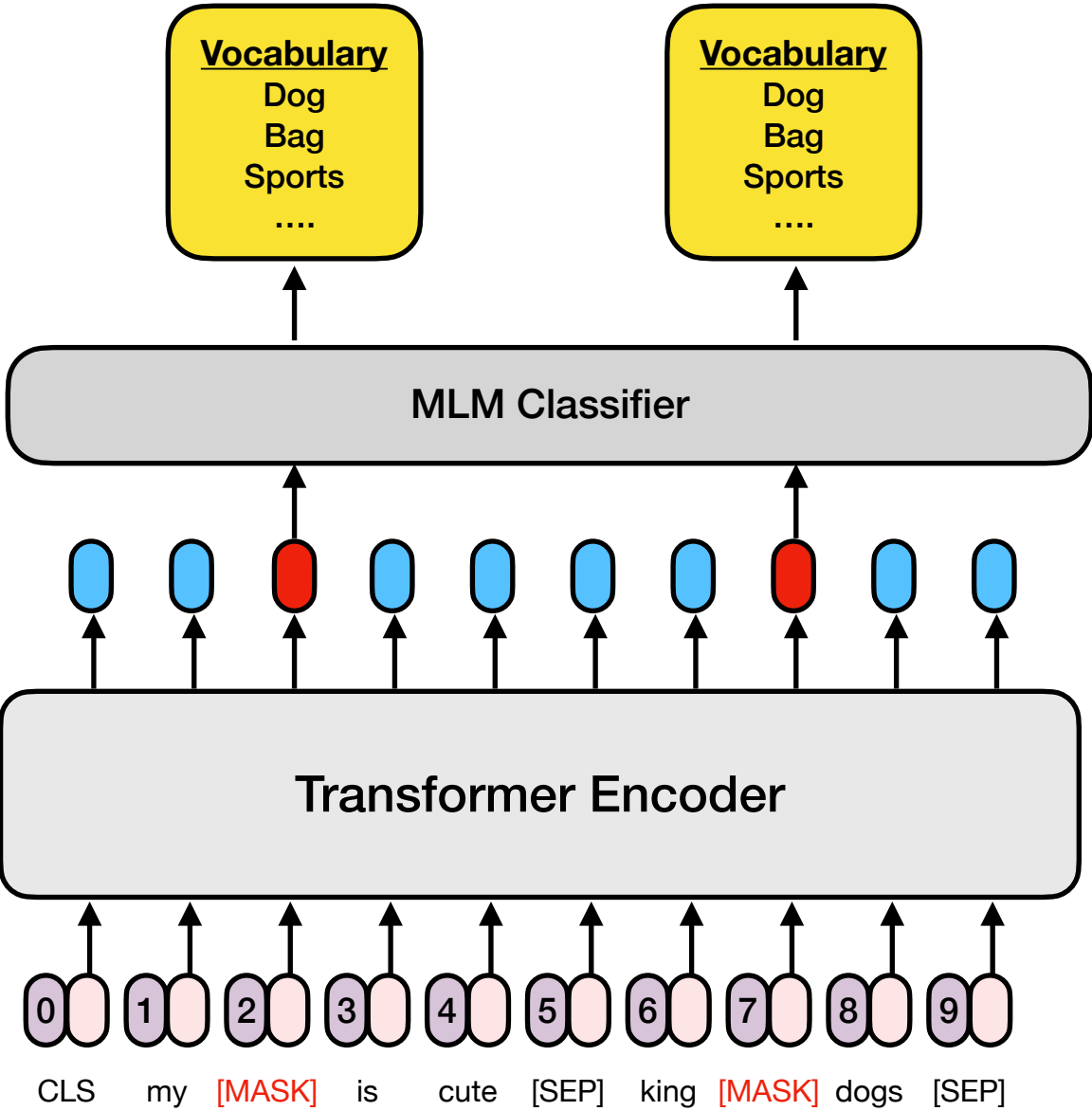
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2022.06.09

- 1. Self-Supervised Learning**
- 2. Token Cut**
- 3. Experiments**
- 4. Laplacian Matrix**

Self-Supervised Learning

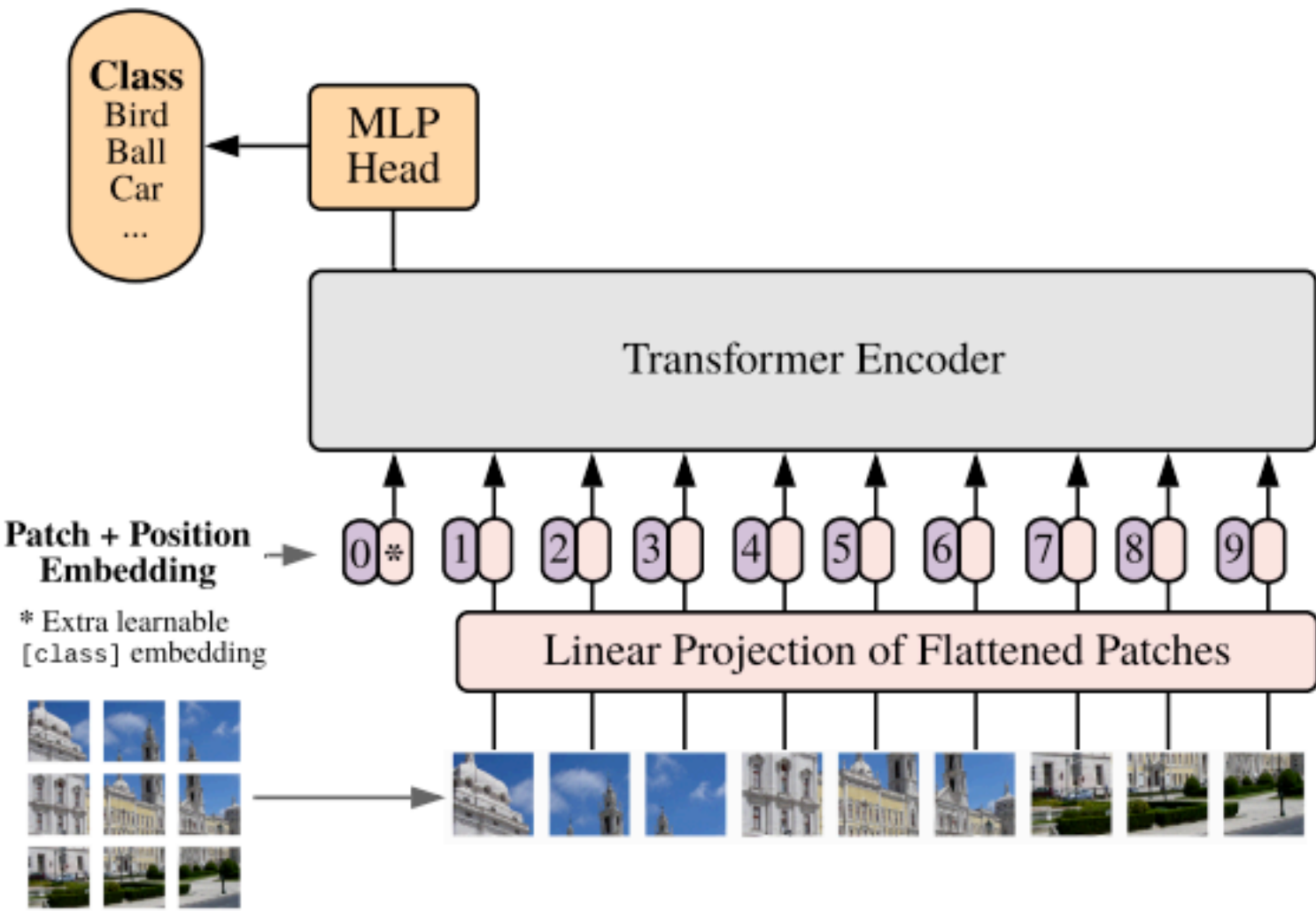
BERT



Unlabeled Text

모든 데이터를 학습 할 수 있음

Vision Transformer(ViT)



Labeled Image

사람이 label 한 데이터만 학습가능

- **Task #1: Image Reconstruction**
- Task #2: Rotation Prediction
- Task #3: Contrastive learning

$$\mathcal{L}_{recons}(W) = \frac{1}{N} \sum_1^N ||x_i - SiT_{recons}(x_i)||$$

- Random Drop
- Random Replace
- Color Distortion
- Blurring
- Grey-scale

Transform Input Image

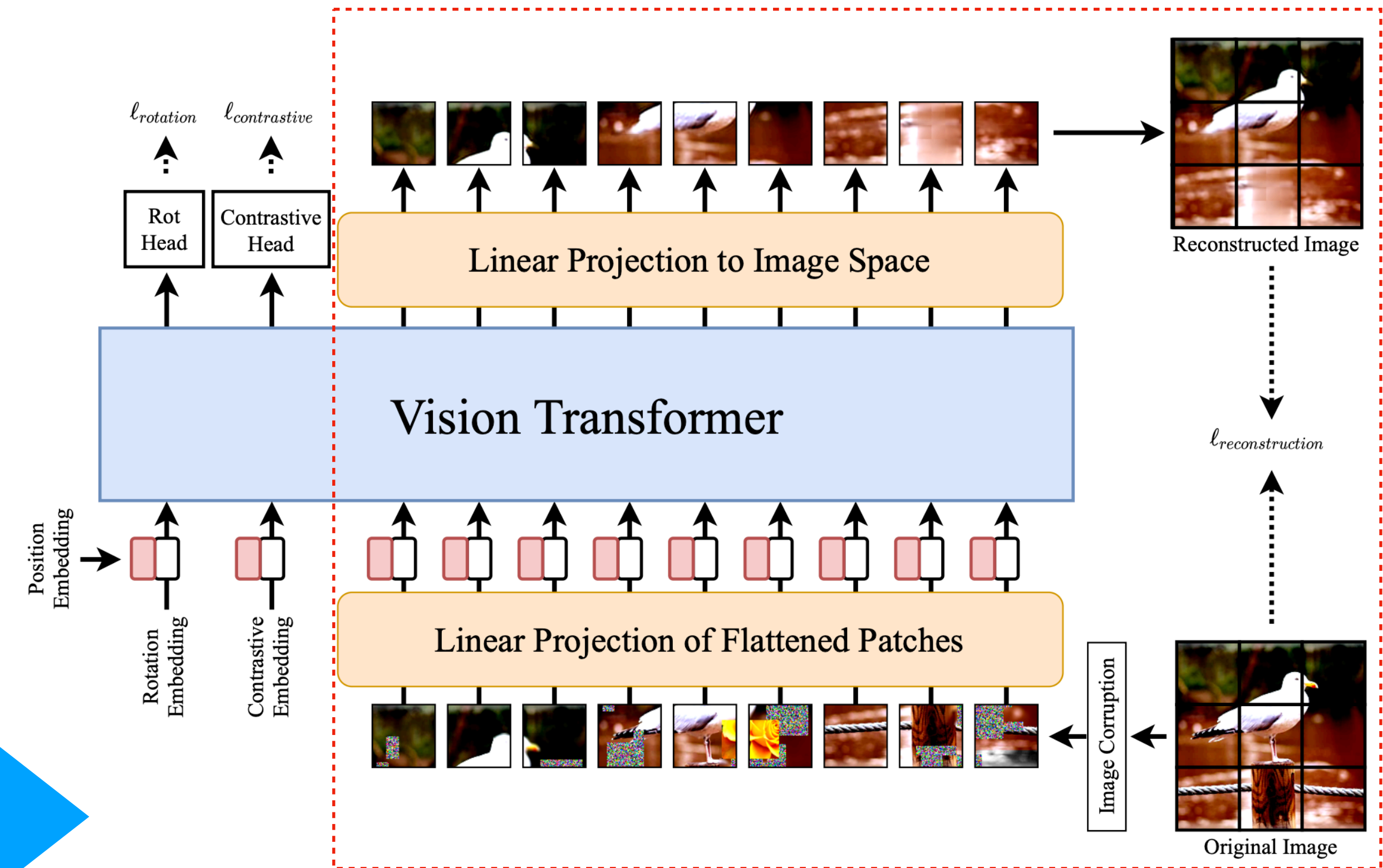
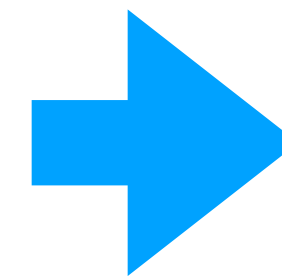


Fig. 1: Self-supervised vision Transformer (SiT)

- Task #1: Image Reconstruction
- **Task #2: Rotation Prediction**
- Task #3: Contrastive learning

$$\hat{y} = \text{softmax}(\text{SiT}_{\text{rotation}}(R(x_i, \theta)))$$

$$\mathcal{L}_{\text{rotation}}(W) = -\frac{1}{N} \sum_i \log \hat{y}^\theta$$

- 0°
- 90°
- 180°
- 270°

Rotate Input Image

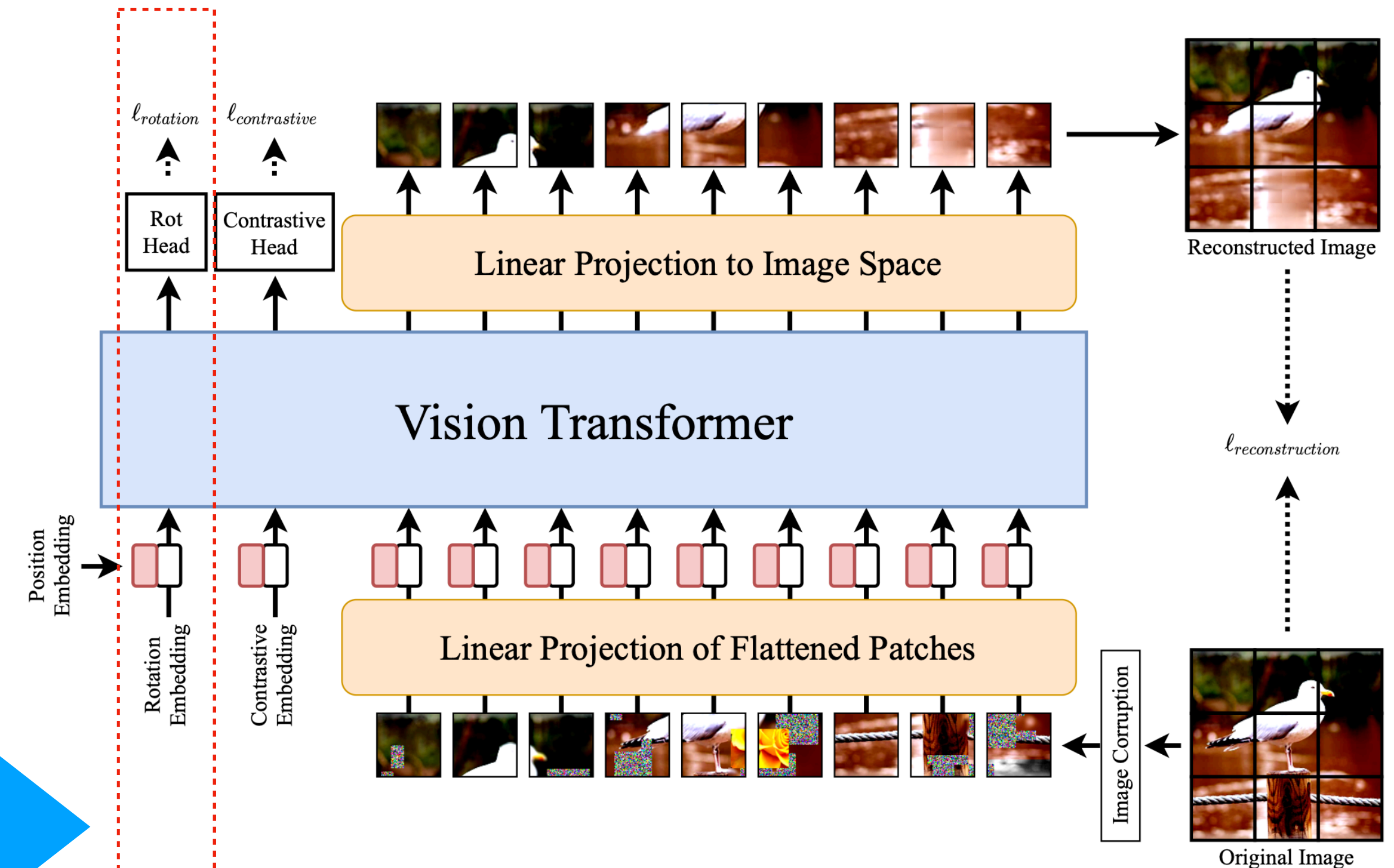
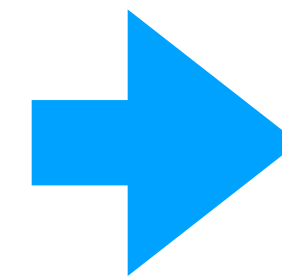


Fig. 1: Self-supervised vIision Transformer (SiT)

- Task #1: Image Reconstruction
- Task #2: Rotation Prediction
- **Task #3: Contrastive learning**

$$l_{contr}^{x_i, x_j} = \frac{e^{\text{sim}(\text{SiT}_{contr}(x_i), \text{SiT}_{contr}(x_j))/\tau}}{\sum_{k=1, k \neq i}^{2N} e^{\text{sim}(\text{SiT}_{contr}(x_i), \text{SiT}_{contr}(x_k))/\tau}}$$

$$\mathcal{L}_{contr}(W) = -\frac{1}{N} \sum_i^N \log l_{contr}^{x_j, x_{\tilde{j}}}(W)$$

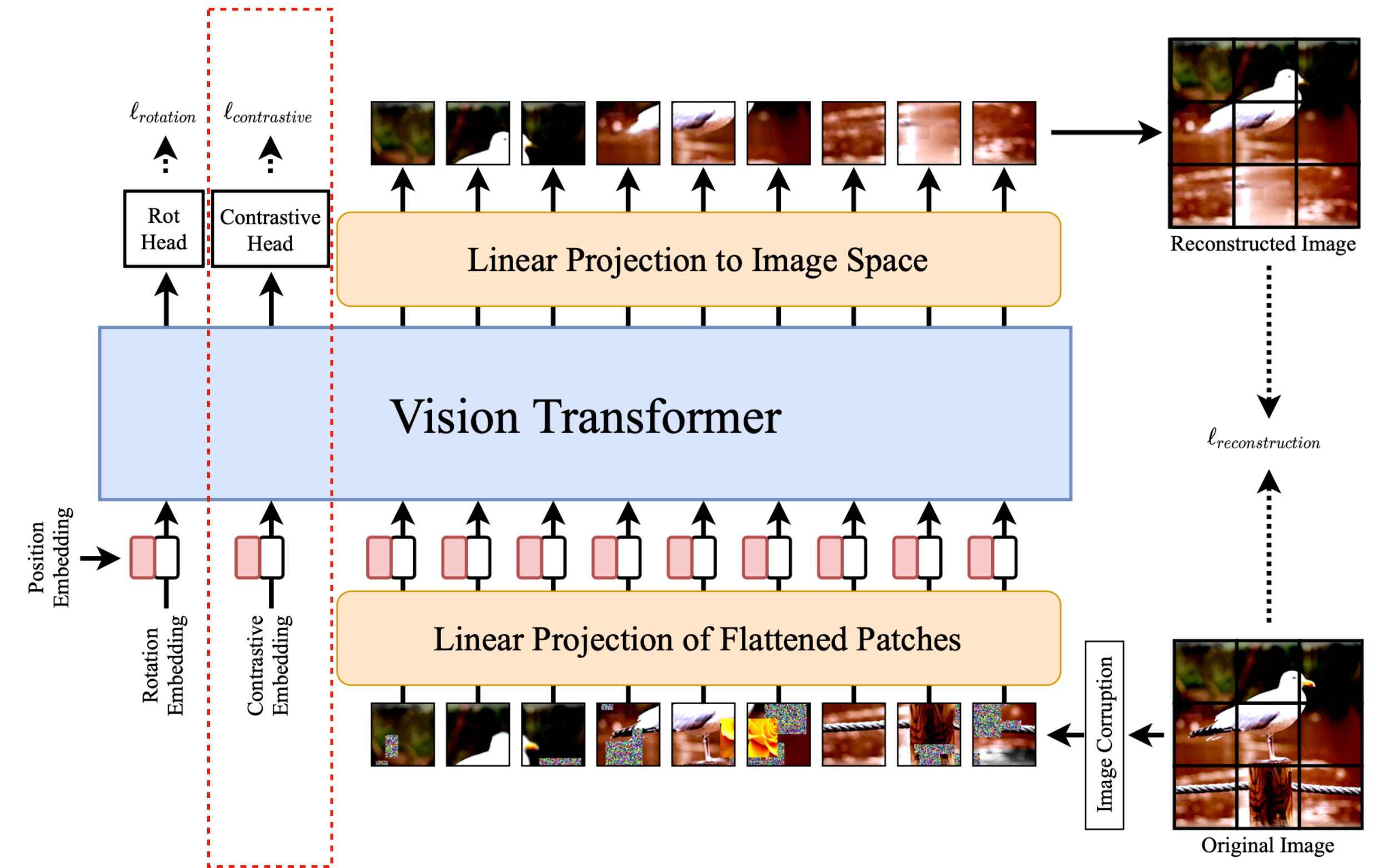


Fig. 1: Self-supervised vIision Transformer (SiT)

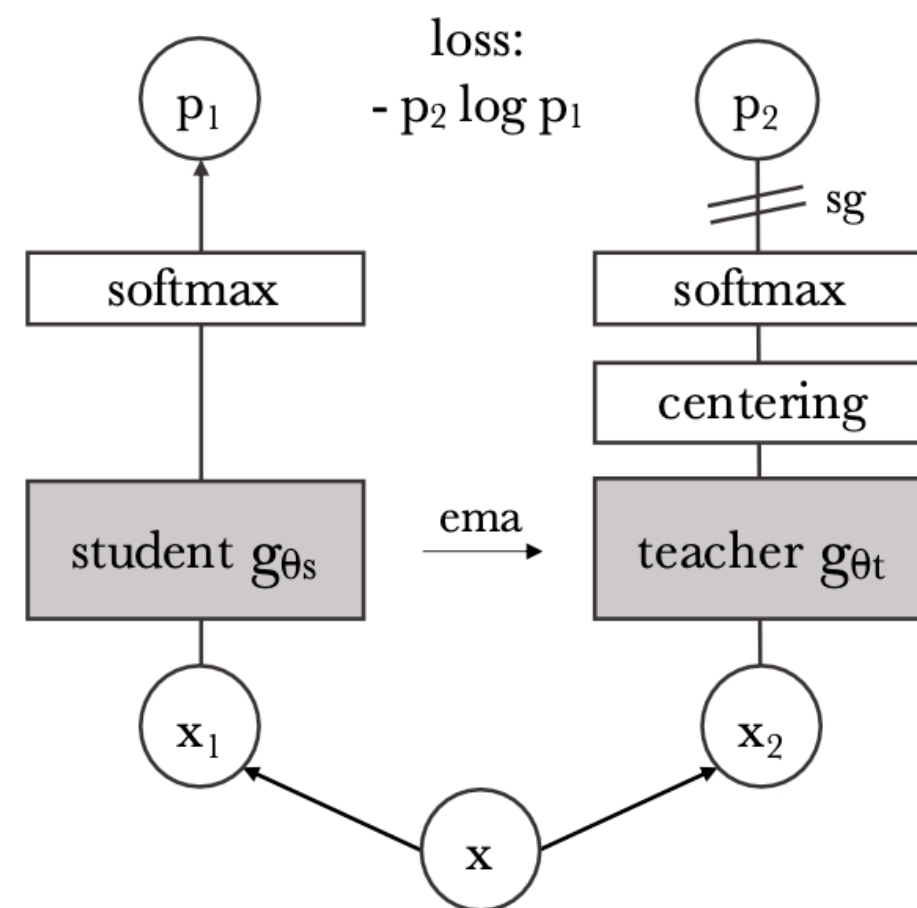
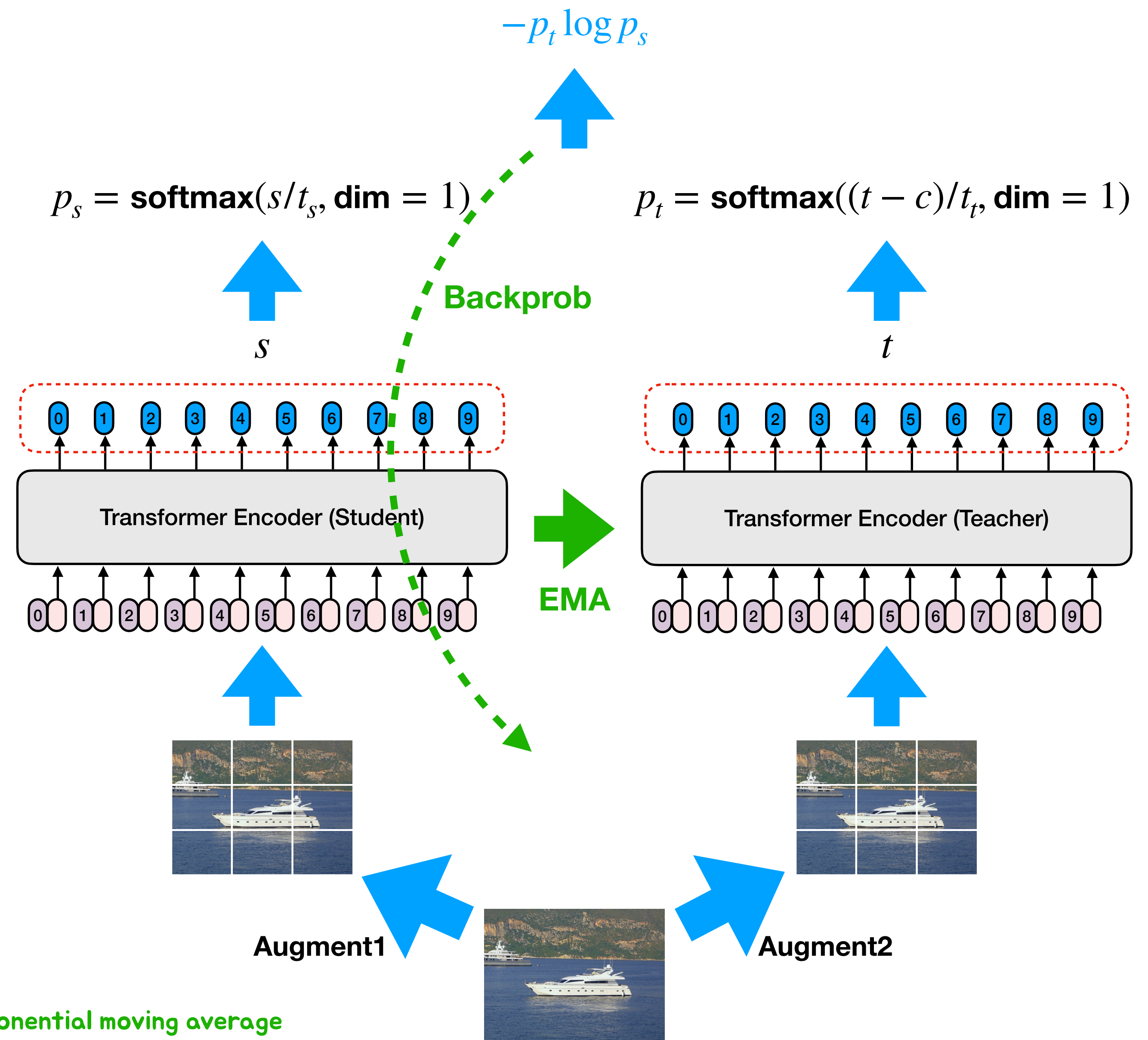


Figure 2: **Self-distillation with no labels.** We illustrate DINO in the case of one single pair of views (x_1, x_2) for simplicity. The model passes two different random transformations of an input image to the student and teacher networks. Both networks have the same architecture but different parameters. The output of the teacher network is centered with a mean computed over the batch. Each networks outputs a K dimensional feature that is normalized with a temperature softmax over the feature dimension. Their similarity is then measured with a cross-entropy loss. We apply a stop-gradient (sg) operator on the teacher to propagate gradients only through the student. The teacher parameters are updated with an exponential moving average (ema) of the student parameters.



Algorithm 1 DINO PyTorch pseudocode w/o multi-crop.

```

# gs, gt: student and teacher networks
# C: center (K)
# tps, tpt: student and teacher temperatures
# l, m: network and center momentum rates
gt.params = gs.params
for x in loader: # load a minibatch x with n samples
    x1, x2 = augment(x), augment(x) # random views

    s1, s2 = gs(x1), gs(x2) # student output n-by-K
    t1, t2 = gt(x1), gt(x2) # teacher output n-by-K

    loss = H(t1, s2)/2 + H(t2, s1)/2
    loss.backward() # back-propagate

    # student, teacher and center updates
    update(gs) # SGD
    gt.params = l*gt.params + (1-l)*gs.params
    C = m*C + (1-m)*cat([t1, t2]).mean(dim=0)

def H(t, s):
    t = t.detach() # stop gradient
    s = softmax(s / tps, dim=1)
    t = softmax((t - C) / tpt, dim=1) # center + sharpen
    return - (t * log(s)).sum(dim=1).mean()
  
```

$$\min_{\theta_s} H(P_t(x), P_s(x)),$$

$$\theta_t \leftarrow \lambda \theta_t + (1 - \lambda) \theta_s \quad \lambda: \text{cosine schedule [0.996, 1]}$$

$$c \leftarrow mc + (1 - m) \frac{1}{B} \sum_{i=1}^B g_{\theta_t}(x_i), \quad \text{Avoiding collapse}$$

$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)} / \tau_s)}{\sum_{k=1}^K \exp(g_{\theta_s}(x)^{(k)} / \tau_s)},$$

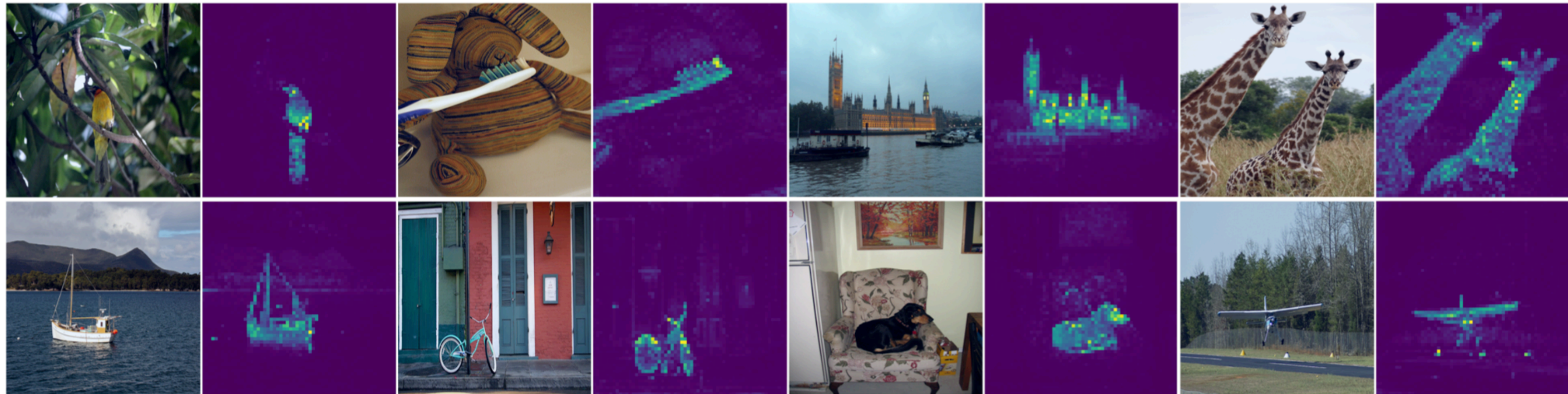


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

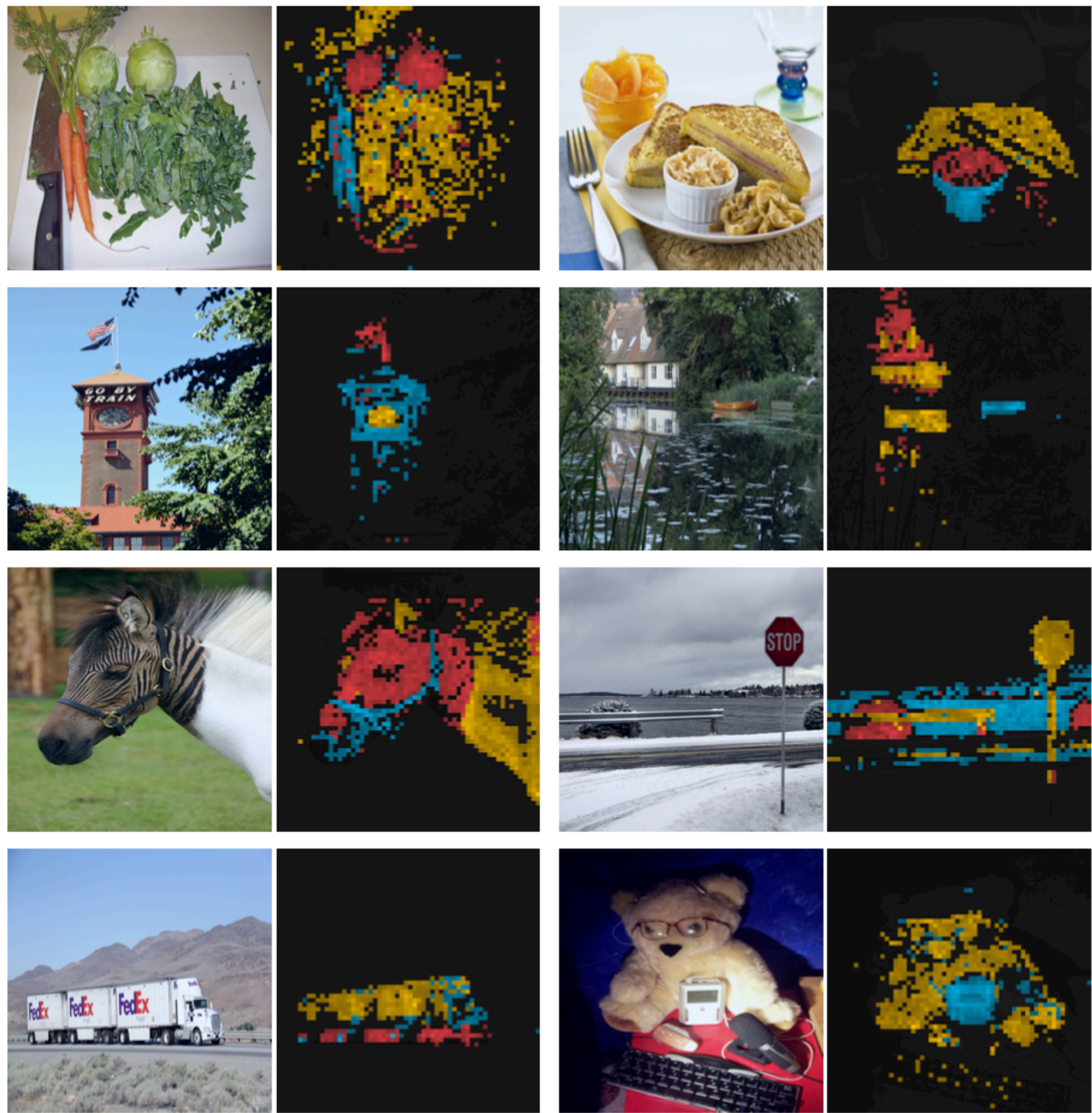
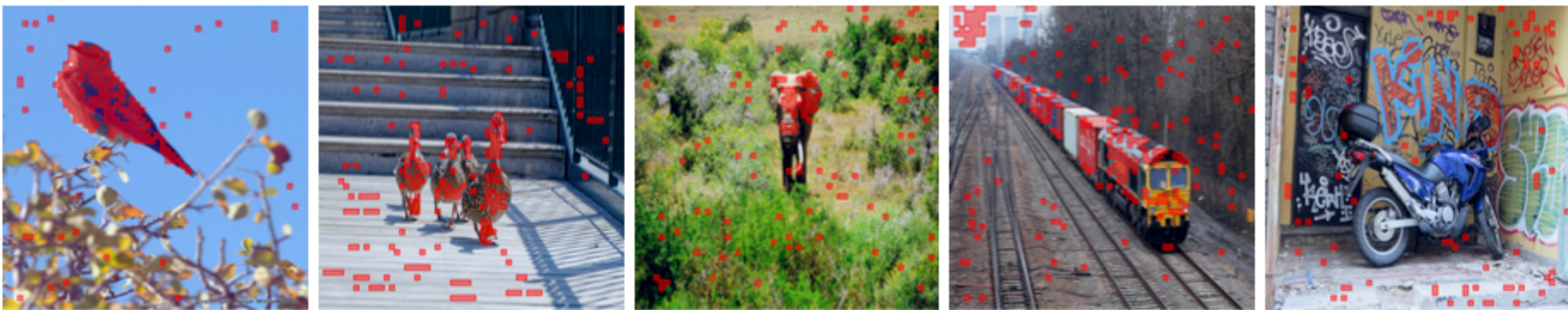


Figure 3: **Attention maps from multiple heads.** We consider the heads from the last layer of a ViT-S/8 trained with DINO and display the self-attention for [CLS] token query. Different heads, materialized by different colors, focus on different locations that represents different objects or parts (more examples in Appendix).

Supervised



DINO



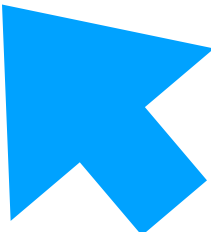
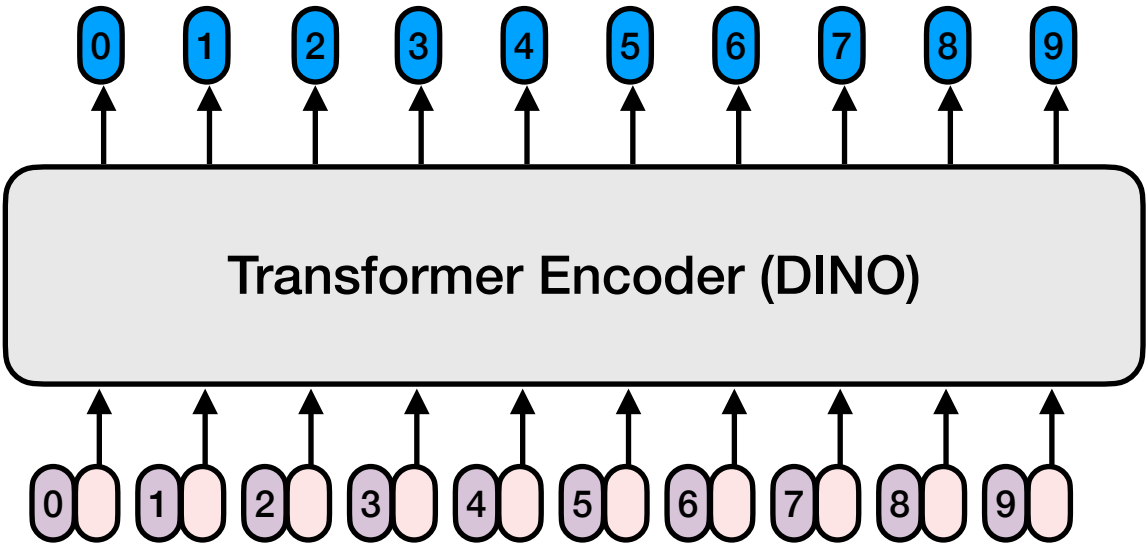
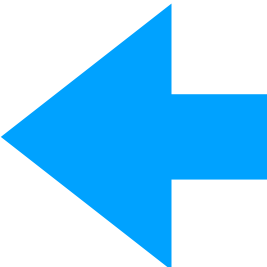
	Random	Supervised	DINO
ViT-S/16	22.0	27.3	45.9
ViT-S/8	21.8	23.7	44.7

Figure 4: **Segmentations from supervised versus DINO.** We visualize masks obtained by thresholding the self-attention maps to keep 60% of the mass. On top, we show the resulting masks for a ViT-S/8 trained with supervision and DINO. We show the best head for both models. The table at the bottom compares the Jac-card similarity between the ground truth and these masks on the validation images of PASCAL VOC12 dataset.

	1	2	3	4	5	6	7	8	9
1	0	1	1	0	0	0	0	0	0
2	1	0	1	0	0	0	0	0	0
3	1	1	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	1	1
5	0	0	0	0	0	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	0	1	0	0	0	1	1
8	0	0	0	1	0	0	1	0	1
9	0	0	0	1	0	0	1	1	0

Adjacency Matrix

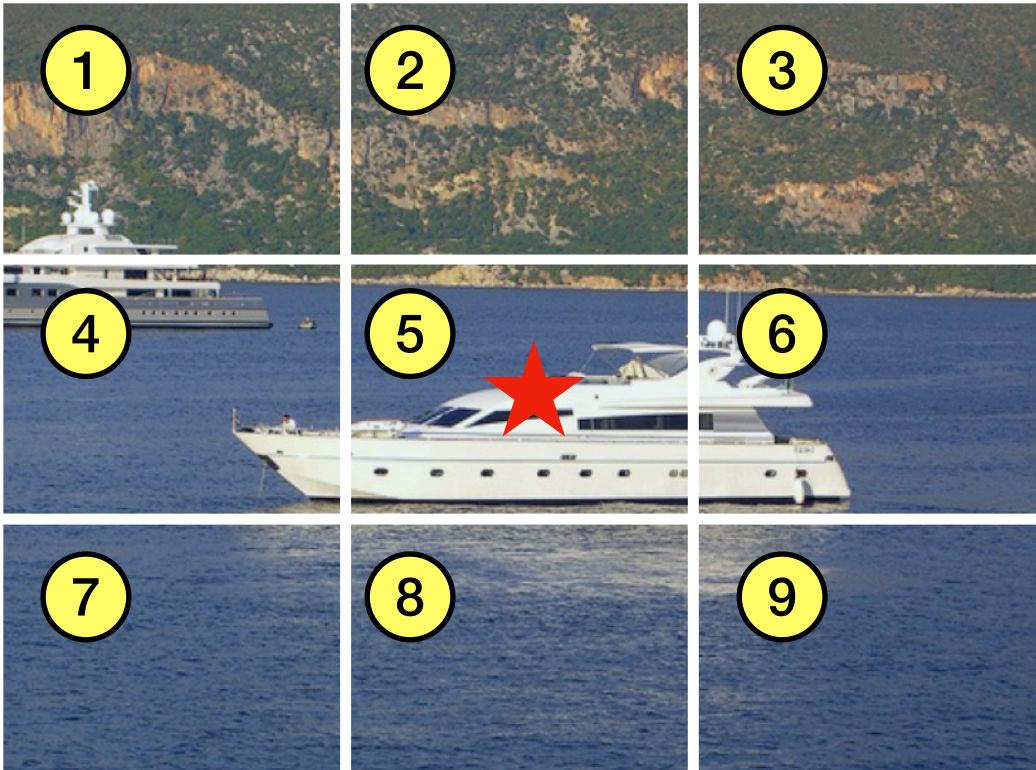
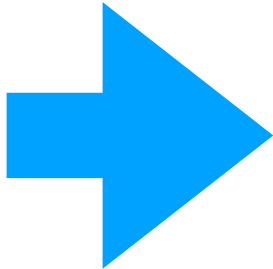
$$a_{pq} = \begin{cases} 1 & \text{if } \mathbf{f}_p^\top \mathbf{f}_q \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$



	1	2	3	4	5	6	7	8	9
1	2	0	0	0	0	0	0	0	0
2	0	2	0	0	0	0	0	0	0
3	0	0	2	0	0	0	0	0	0
4	0	0	0	4	0	0	0	0	0
5	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	2	0	0	0
7	0	0	0	0	0	0	3	0	0
8	0	0	0	0	0	0	0	3	0
9	0	0	0	0	0	0	0	0	3

Degree Matrix

seed: smallest degree



	1	2	3	4	5	6	7	8	9
1	0	1	1	0	0	0	0	0	0
2	1	0	1	0	0	0	0	0	0
3	1	1	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	1	1
5	0	0	0	0	0	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	0	1	0	0	0	1	1
8	0	0	0	1	0	0	1	0	1
9	0	0	0	1	0	0	1	1	0

Adjacency Matrix

	1	2	3	4	5	6	7	8	9
1	2	0	0	0	0	0	0	0	0
2	0	2	0	0	0	0	0	0	0
3	0	0	2	0	0	0	0	0	0
4	0	0	0	4	0	0	0	0	0
5	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	2	0	0	0
7	0	0	0	0	0	0	3	0	0
8	0	0	0	0	0	0	0	3	0
9	0	0	0	0	0	0	0	0	3

Degree Matrix

Seed

5

Seed Expansion

$$\mathcal{S} = \{q \mid q \in \mathcal{D}_k \text{ and } \mathbf{f}_q^\top \mathbf{f}_{p^*} \geq 0\}$$

5

6

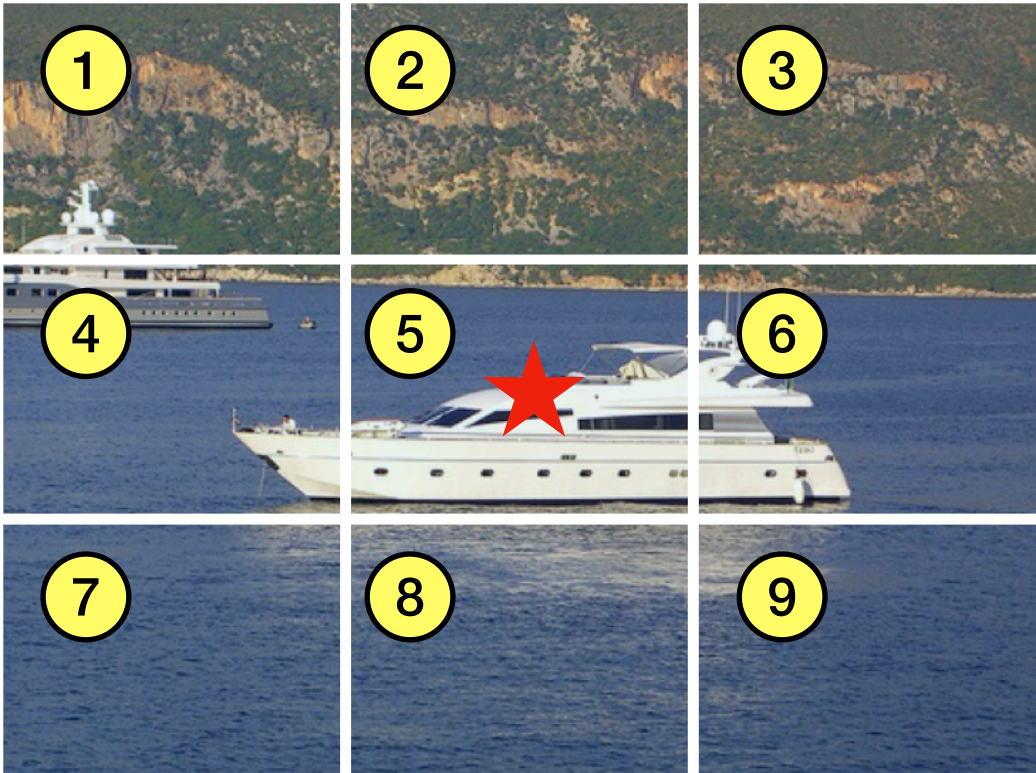
Box Extraction

$$m_q = \begin{cases} 1 & \text{if } \sum_{s \in \mathcal{S}} \mathbf{f}_q^\top \mathbf{f}_s \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

4

5

6



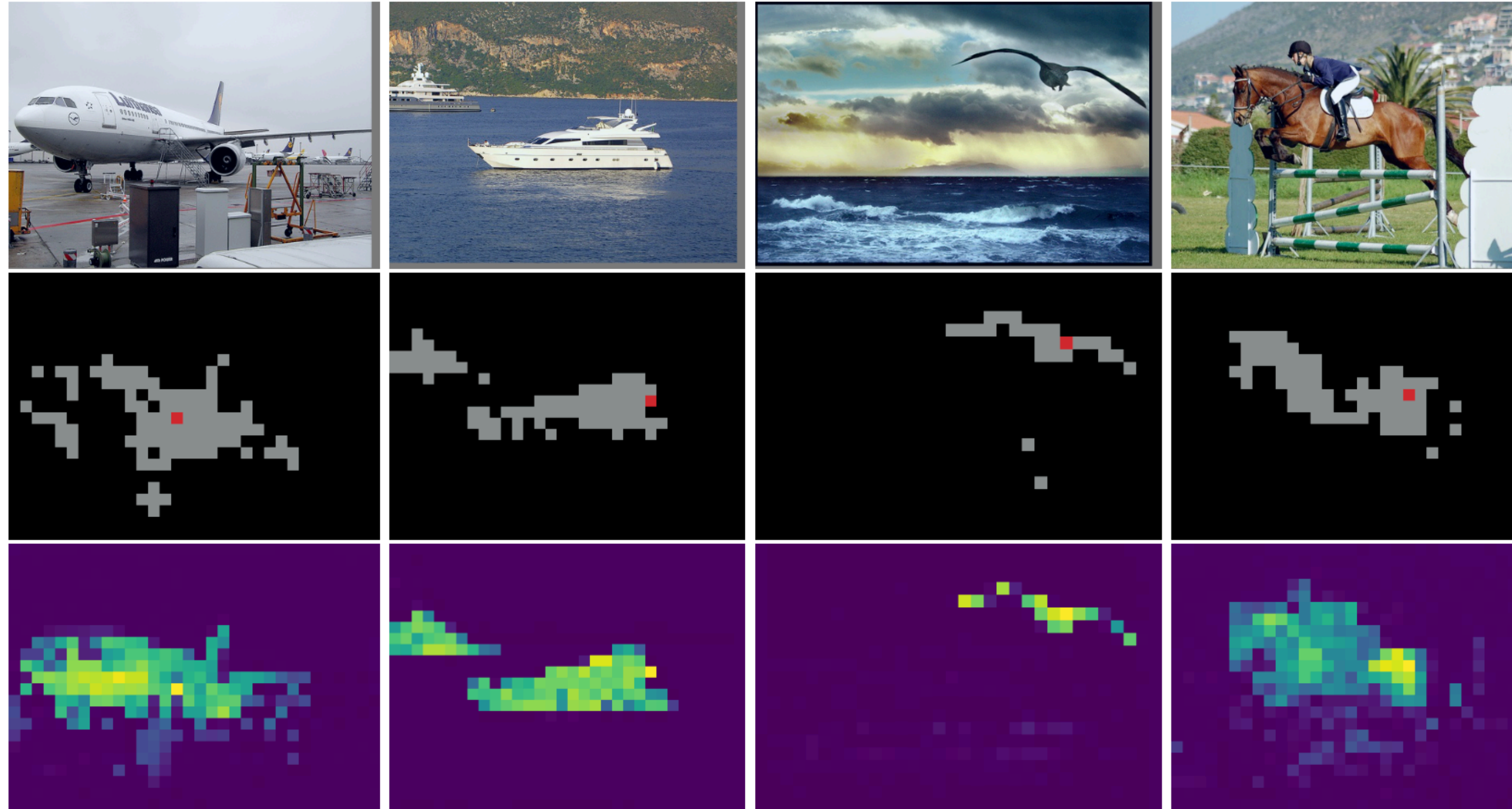


Figure 2: **Initial seed, patch similarities and patch degrees.** Top: images from Pascal VOC2007. Middle: initial seed p^* (in red) and patches similar to p^* (in grey), *i.e.*, such that $\mathbf{f}_p^\top \mathbf{f}_q \geq 0$ hence $a_{p^*q} = 1$. Bottom: map of inverse degrees $1/d_p$ of all patches p (yellow to blue, for low to high degrees). The initial seed p^* is the patch with the lowest degree. Figure is best viewed in color.

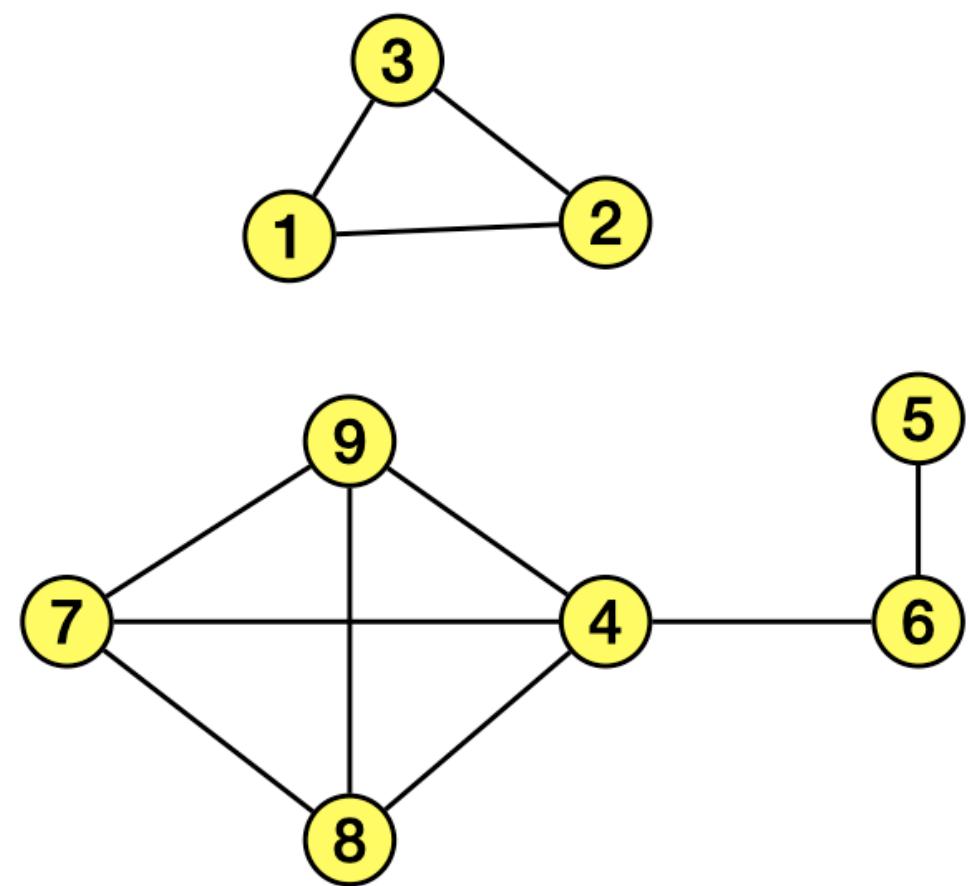
Token Cut

	1	2	3	4	5	6	7	8	9
1	0	1	1	0	0	0	0	0	0
2	1	0	1	0	0	0	0	0	0
3	1	1	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	1	1
5	0	0	0	0	0	1	0	0	0
6	0	0	0	1	1	0	0	0	0
7	0	0	0	1	0	0	0	1	1
8	0	0	0	1	0	0	1	0	1
9	0	0	0	1	0	0	1	1	0

Adjacency Matrix

	1	2	3	4	5	6	7	8	9
1	2	0	0	0	0	0	0	0	0
2	0	2	0	0	0	0	0	0	0
3	0	0	2	0	0	0	0	0	0
4	0	0	0	4	0	0	0	0	0
5	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	2	0	0	0
7	0	0	0	0	0	0	3	0	0
8	0	0	0	0	0	0	0	3	0
9	0	0	0	0	0	0	0	0	3

Degree Matrix



$$\mathcal{E}_{i,j} = \begin{cases} 1, & \text{if } S(v_i, v_j) \geq \tau \\ \epsilon, & \text{else} \end{cases}$$

$$S(v_i, v_j) = \frac{v_i v_j}{\|v_i\|_2 \|v_j\|_2}.$$

$\tau : 0.2$
 $\epsilon : 1e-5$

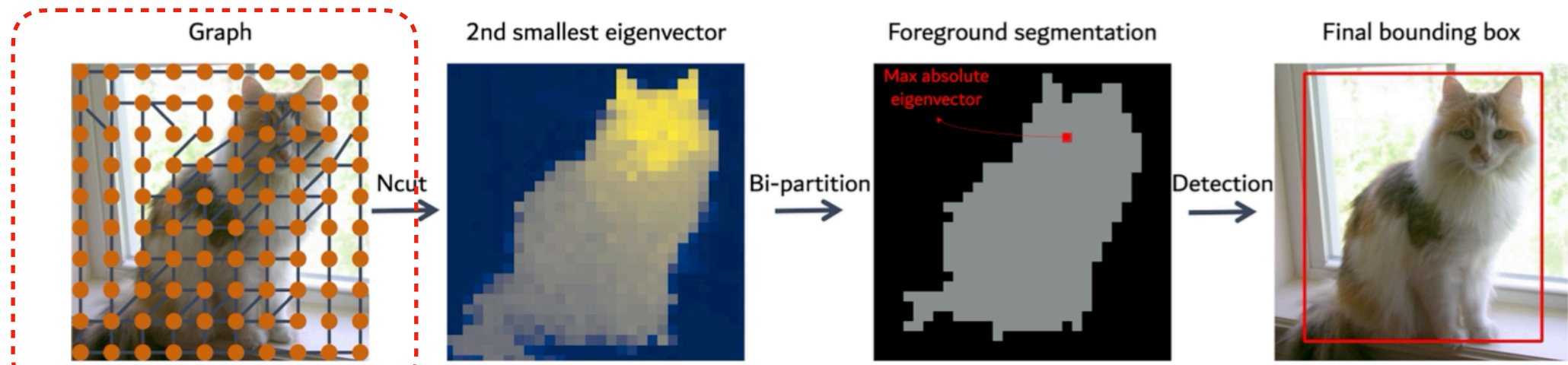
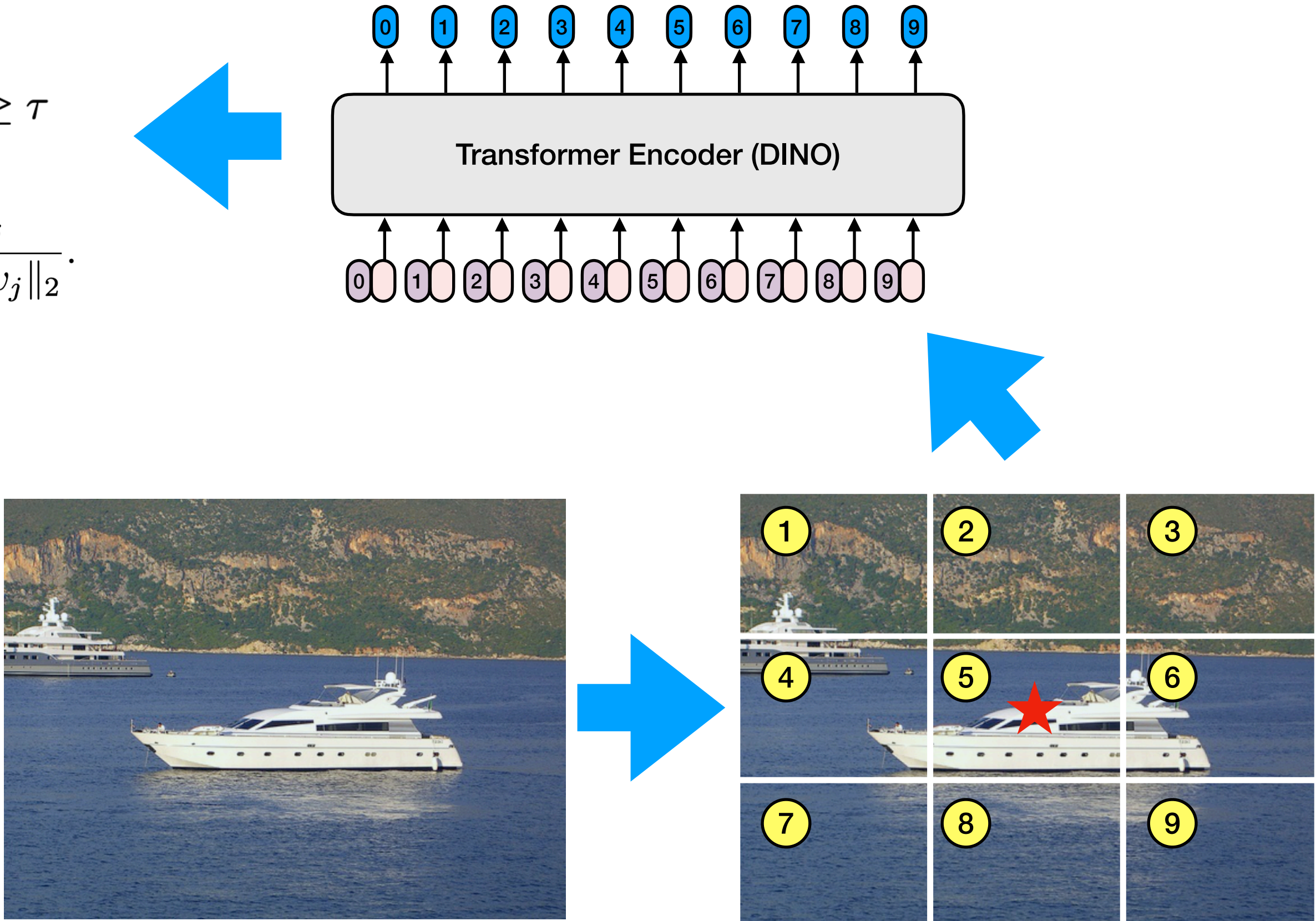
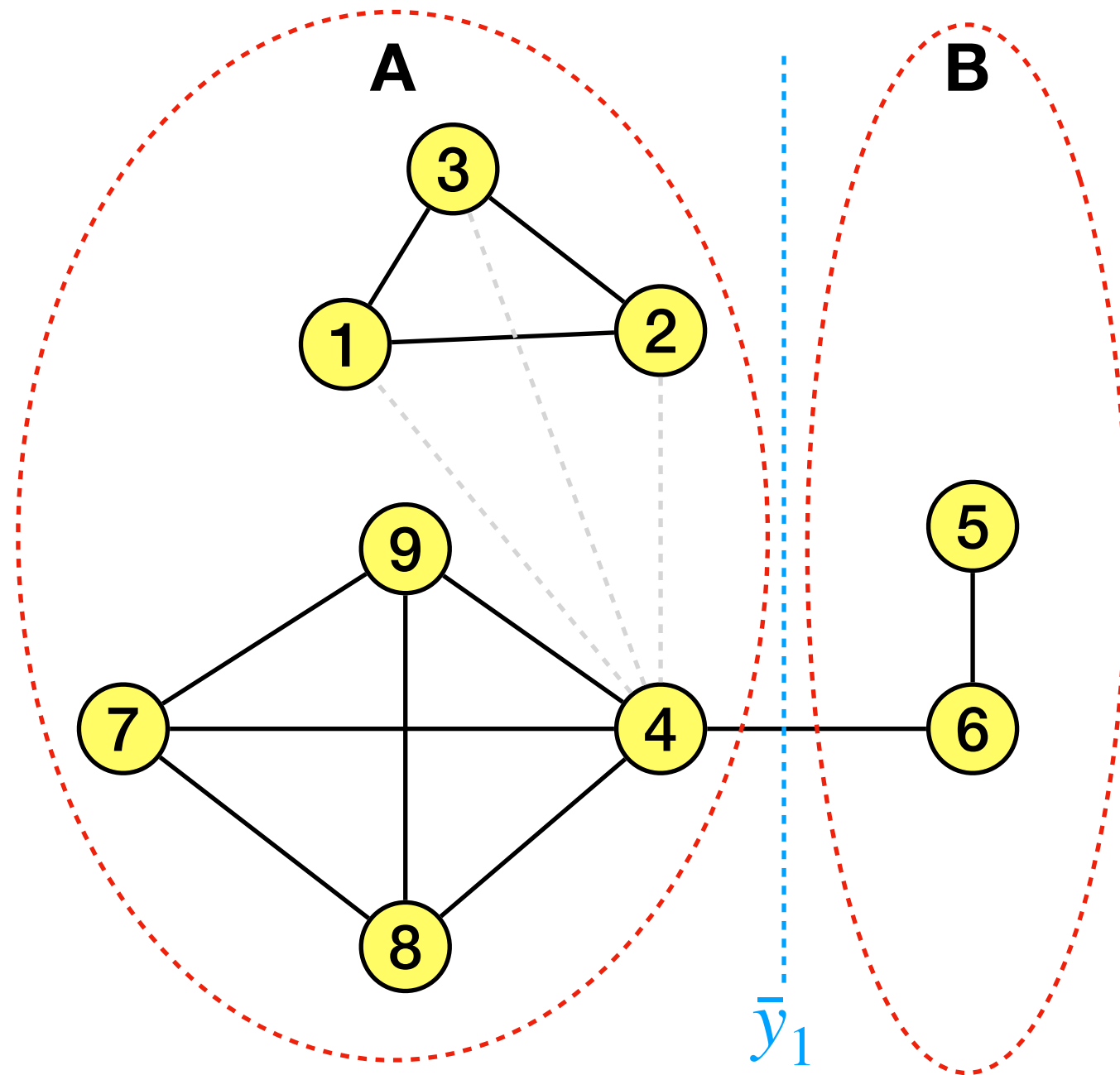


Figure 2: An overview of the TokenCut approach. We construct a graph where the nodes are tokens and the edges are similarities between the tokens using transformer features. The foreground and background segmentation can be solved by Ncut [43]. Performing bi-partition on the second smallest eigenvector allows to detect foreground object.





$\lambda_2 : 2_{nd}$ **smallest eigen value & vector**

$y = \{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$

$$\bar{y}_1 = \frac{1}{N} \sum_i y_1^i$$

$$A = \{v_i | y_1^i \leq \bar{y}_1\}$$

$$B = \{v_i | y_1^i > \bar{y}_1\}$$

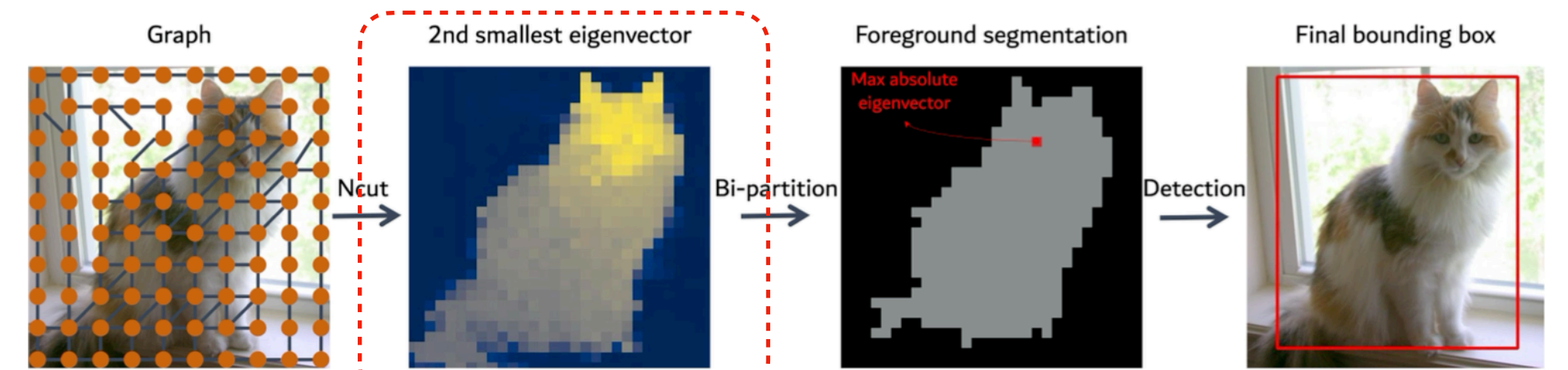
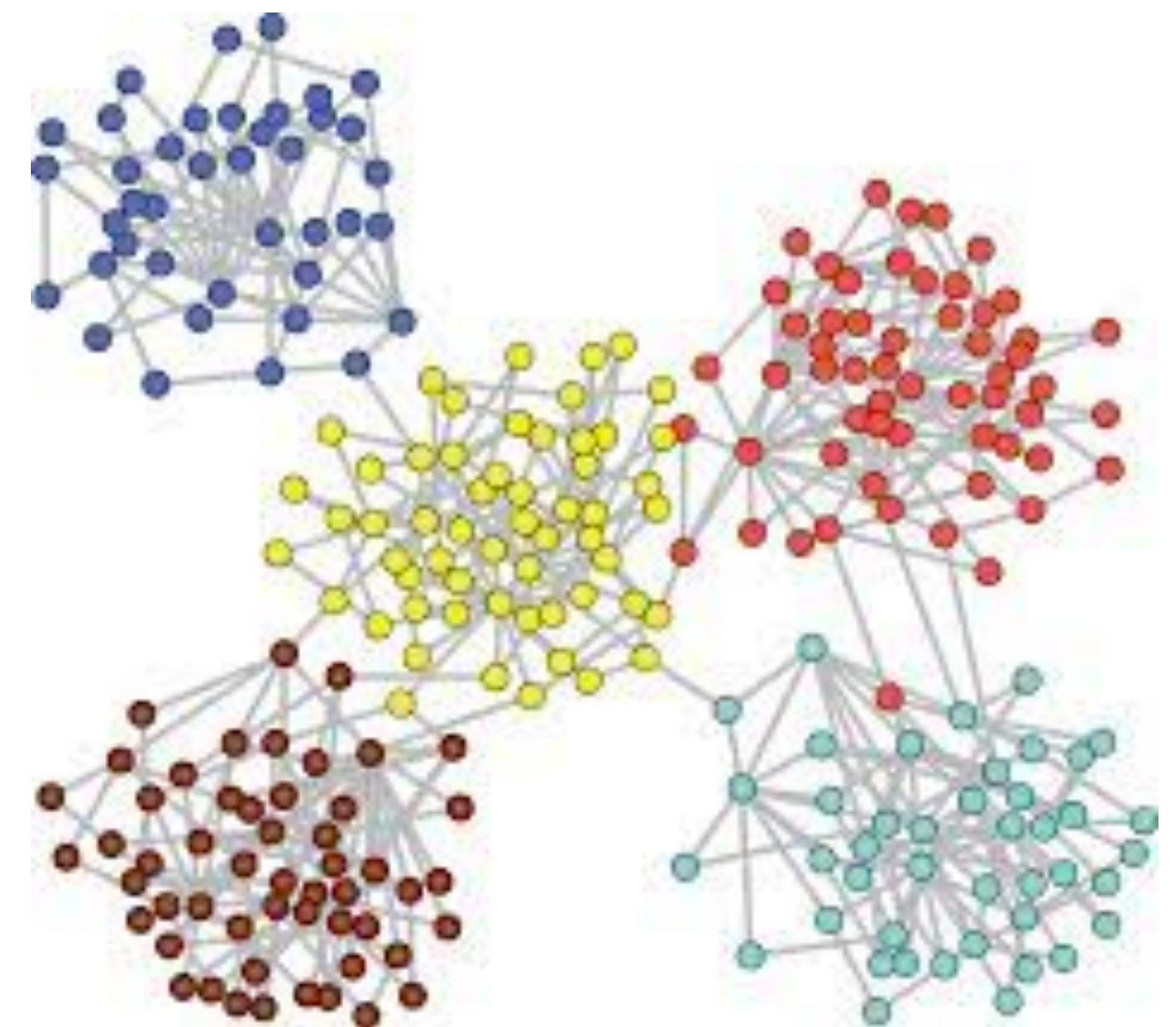
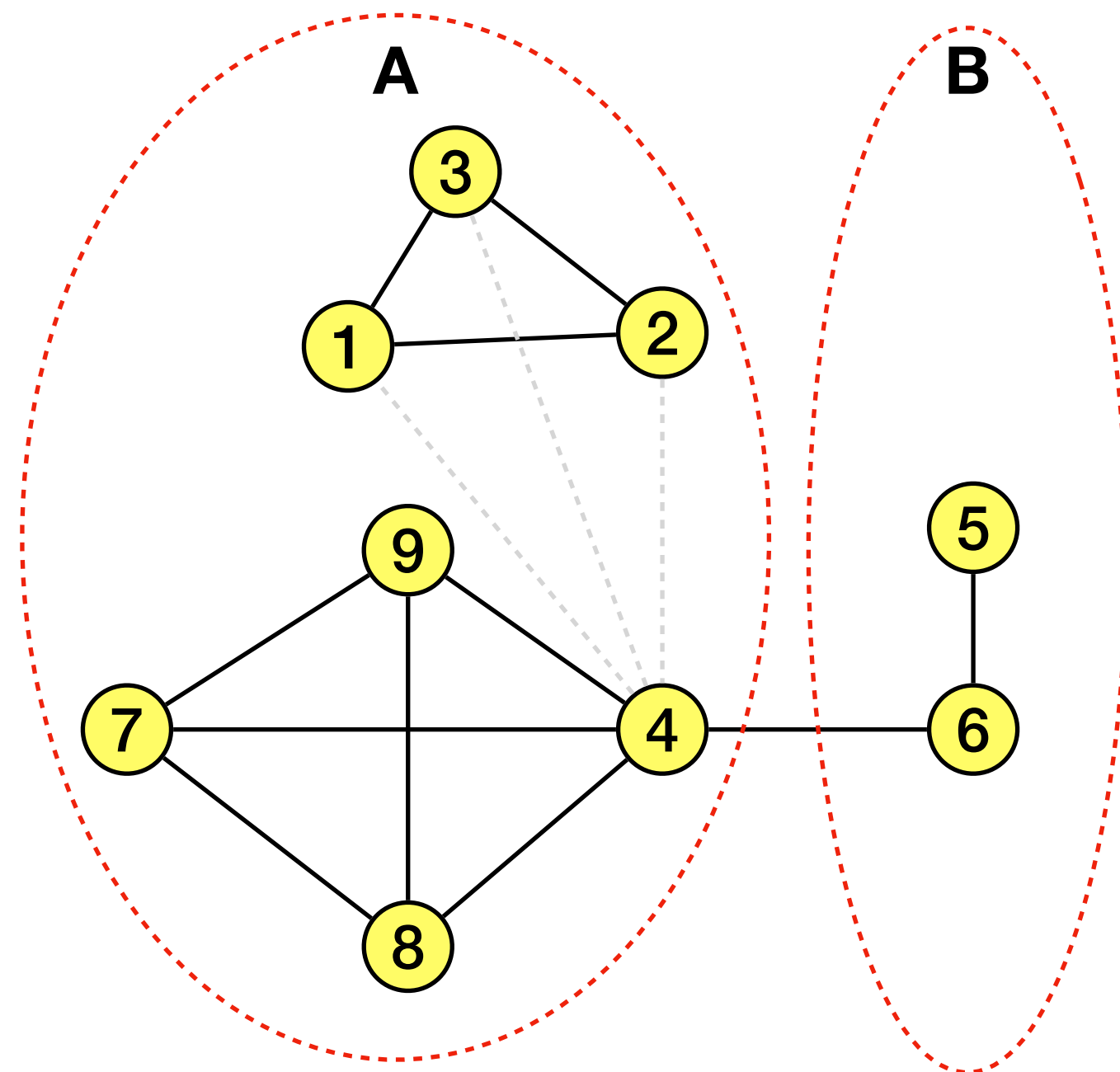


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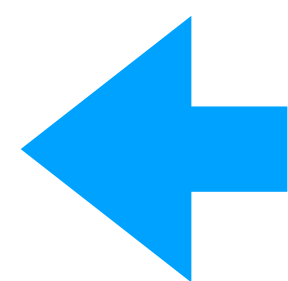




$\lambda_2 : 2_{nd}$ smallest eigen value & vector

$y = \{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$

v_{max}



y_i 의 절대값이 최대인 위치 ⑤
가 속한곳 foreground

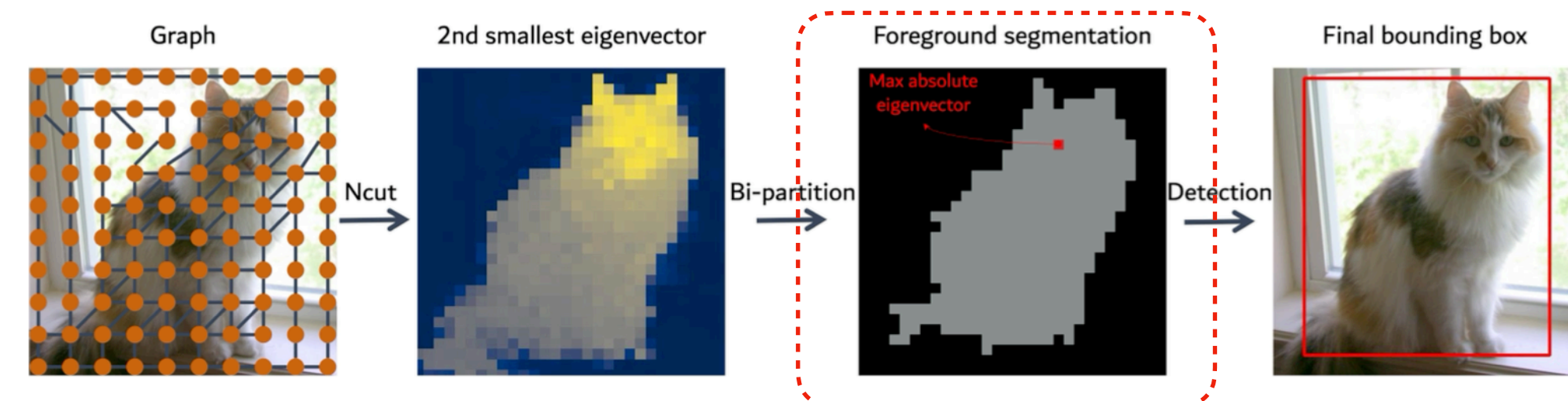


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select the **largest connected component existing in the foreground containing the maximum absolute value v_{max}** as our final object region.

Experiments

- Evaluation metric

- CorLoc**: IoU가 0.5 이상인 bounding box 개수

$$\frac{TP}{TP + FP}$$

Table 1: **Comparisons for unsupervised single object discovery.** We compare TokenCut to state-of-the-art object discovery methods on VOC07 [19], VOC12 [20] and COCO20K [33, 56] datasets. Model performances are evaluated with CorLoc metric. “Inter-image Simi.” means the model leverages information from the entire dataset and explore inter-image similarities to localize objects.

Method	Inter-image Simi.	DINO [6] Feat.	VOC07 [19]	VOC12 [20]	COCO20K [33, 56]
Selective Search [45, 51]	✗	-	18.8	20.9	16.0
EdgeBoxes [45, 79]	✗	-	31.1	31.6	28.8
Kim et al. [30, 45]	✓	-	43.9	46.4	35.1
Zhang et al. [45, 74]	✓	-	46.2	50.5	34.8
DDT+ [45, 61]	✓	-	50.2	53.1	38.2
rOSD [45, 56]	✓	-	54.5	55.3	48.5
LOD [45, 57]	✓	-	53.6	55.1	48.5
DINO-seg [6, 45]	✗	ViT-S/16 [18]	45.8	46.2	42.1
LOST [45]	✗	ViT-S/16 [18]	61.9	64.0	50.7
TokenCut	✗	ViT-S/16 [18]	68.8 (↑ 6.9)	72.1 (↑ 8.1)	58.8 (↑ 8.1)
LOD + CAD* [45]	✓	-	56.3	61.6	52.7
rOSD + CAD* [45]	✓	-	58.3	62.3	53.0
LOST + CAD* [45]	✗	ViT-S/16 [18]	65.7	70.4	57.5
TokenCut + CAD* [45]	✗	ViT-S/16 [18]	71.4 (↑ 5.7)	75.3 (↑ 4.9)	62.6 (↑ 5.1)

* +CAD indicates to train a second stage class-agnostic detector with “pseudo-boxes” labels.

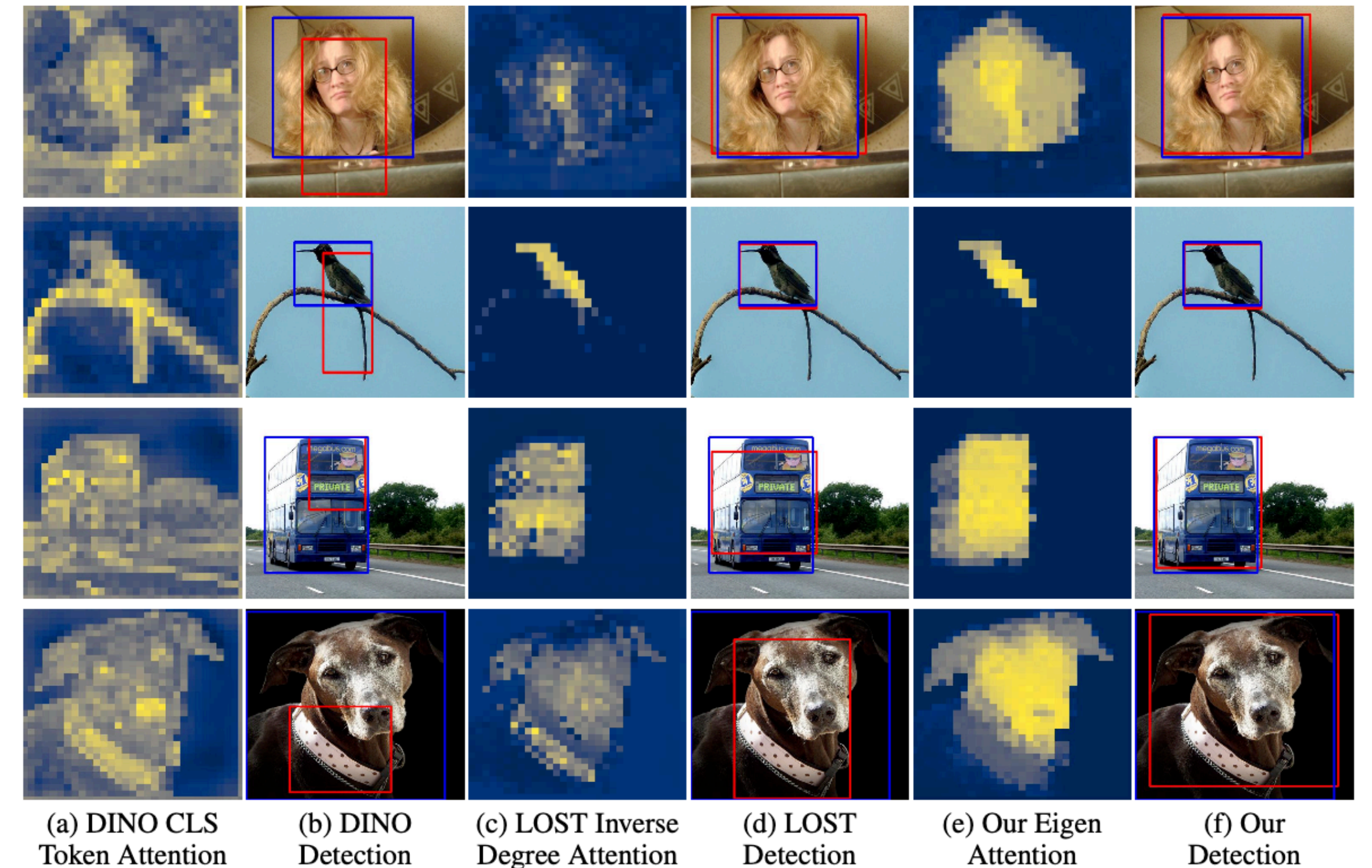


Figure 3: **Visual results of unsupervised single object discovery on VOC12.** In (a), we show the attention of the CLS token in DINO [6] which is used for detection (b). LOST [45] is mainly relied on the map of inverse degrees (c) to perform detection (d). For our approach, we illustrate the eigenvector in (e) and our detection in (f). Blue and Red bounding boxes indicate the ground-truth and the predicted bounding boxes respectively.

- Evaluation metric
 - **Top-1 Cls**: top accuracy of image classification.
 - **GT Loc**: IoU가 0.5 이상인 bounding box 개수
 - **Top-1 Loc**: (Top-1CLS + GT Loc) image class가 정답이고 IoU가 0.5 이상인 경우

Table 2: **Comparisons for weakly supervised object localization.** We report Top-1 Cls, GT Loc and Top-1 Loc on CUB [59] and ImageNet-1K [14] datasets. Compared state-of-the-art methods are divided into two groups: with ImageNet-1K *supervised* pretraining and with ImageNet-1K *self-supervised* pretraining.

Pretrained Dataset	Method	Backbone	CUB [59], Acc. (%)			ImageNet-1K [14], Acc. (%)		
			Top-1 Cls	GT Loc	Top-1 Loc	Top-1 Cls	GT Loc	Top-1 Loc
ImageNet-1K [14] supervised pretrain	CAM [77]	GoogLeNet [48]	73.8	-	41.1	65	-	43.6
	HaS-32 [46] + [3]	GoogLeNet [48]	75.4	61.1	47.4	68.9	60.6	44.6
	ADL [13] + [3]	ResNet50 [24]	75.0	77.6	59.5	75.8	62.2	49.4
	ADL [13]	InceptionV3 [49]	74.6	-	53.0	72.8	-	48.7
	I2C [76]	InceptionV3 [49]	-	72.6	56	73.3	68.5	53.1
	PSOL [71] [‡]	InceptionV3 [49]	-	-	65.5	-	65.2	54.8
	SLT-Net [22] [★]	InceptionV3 [49]	76.4	86.5	66.1	78.1	67.6	55.7
ImageNet-1K [14] self-supervised pretrain	LOST [45]	ViT-S/16 [18]	79.5	89.7	71.3	77.0	60.0	49
	TokenCut	ViT-S/16 [18]	79.5	91.8 (↑ 2.1)	72.9 (↑ 1.6)	77.0	65.4 (↑ 5.4)	52.3 (↑ 3.3)

★ uses ten crop augmentations to get final classification results. ★ and ‡ learn a classifier and a detector separately.

- Evaluation metric
 - F-measure
 - IoU
 - Accuracy

Table 3: **Comparisons for unsupervised saliency detection** We compare TokenCut to state-of-the-art unsupervised saliency detection methods on ECSSD [44], DUTS [60] and DUT-OMRON [67]. TokenCut achieves better results comparing with other competitive approaches.

Method	ECSSD [44]			DUTS [60]			DUT-OMRON [67]		
	$maxF_{\beta}(\%)$	IoU(%)	Acc.(%)	$maxF_{\beta}(\%)$	IoU(%)	Acc.(%)	$maxF_{\beta}(\%)$	IoU(%)	Acc.(%)
HS [66]	67.3	50.8	84.7	50.4	36.9	82.6	56.1	43.3	84.3
wCtr [78]	68.4	51.7	86.2	52.2	39.2	83.5	54.1	41.6	83.8
WSC [31]	68.3	49.8	85.2	52.8	38.4	86.2	52.3	38.7	86.5
DeepUSPS [37]	58.4	44.0	79.5	42.5	30.5	77.3	41.4	30.5	77.9
BigBiGAN [58]	78.2	67.2	89.9	60.8	49.8	87.8	54.9	45.3	85.6
E-BigBiGAN [58]	79.7	68.4	90.6	62.4	51.1	88.2	56.3	46.4	86.0
LOST [42, 45]	75.8	65.4	89.5	61.1	51.8	87.1	47.3	41.0	79.7
LOST [42, 45]+Bilateral Solver [5]	83.7	72.3	91.6	69.7	57.2	88.7	57.8	48.9	81.8
TokenCut	80.3	71.2	91.8	67.2	57.6	90.3	60.0	53.3	88.0
TokenCut + Bilateral Solver [5]	87.4 (↑ 3.7)	77.2 (↑ 4.9)	93.4 (↑ 1.8)	75.5 (↑ 5.8)	62.4 (↑ 5.2)	91.4 (↑ 2.7)	69.7 (↑ 11.9)	61.8 (↑ 12.9)	89.7 (↑ 7.9)

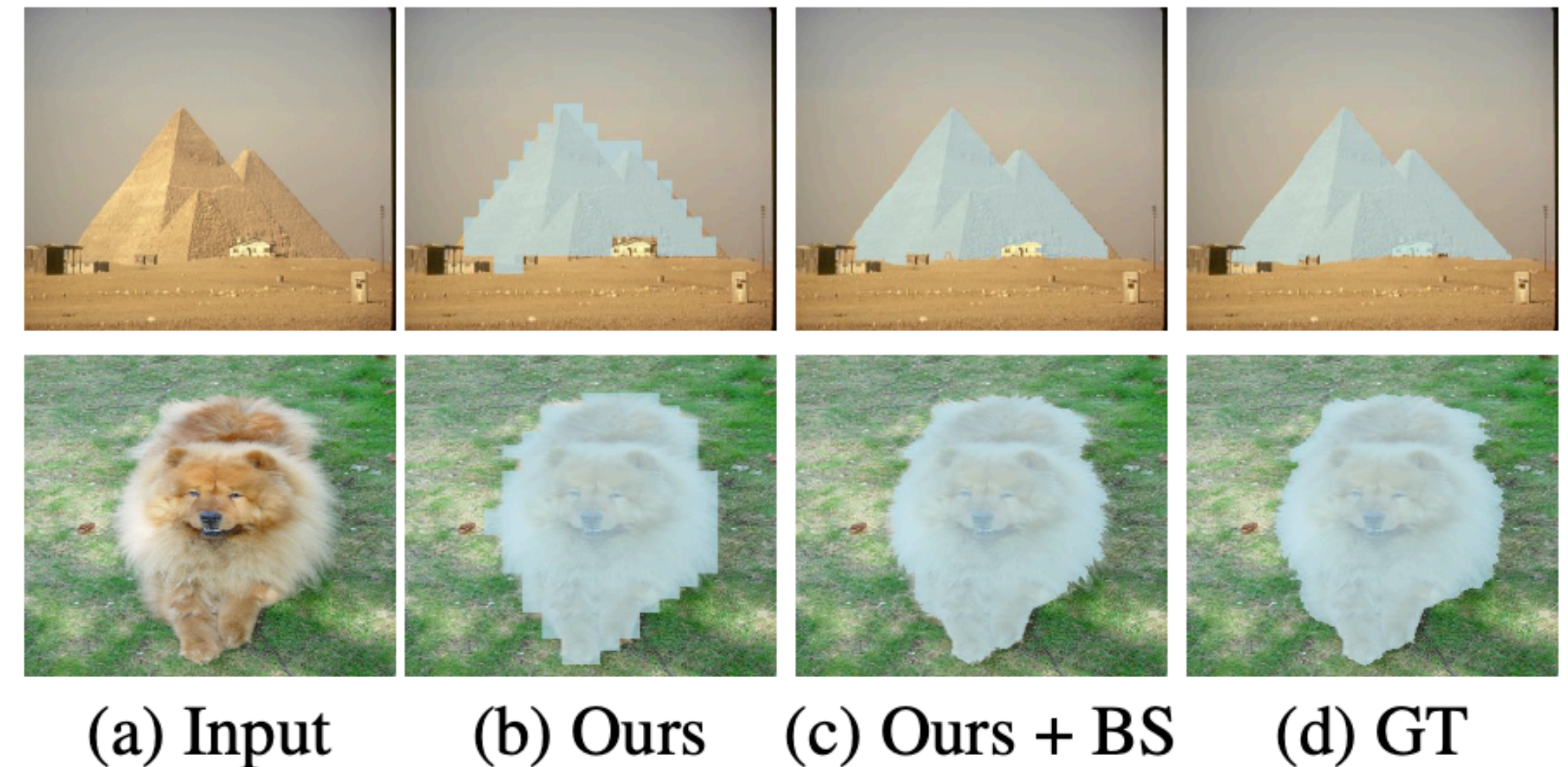


Figure 4: **Visual results of unsupervised segments on ECSSD [44]**. In (a), we show the input image. TokenCut detection result is presented in (b). TokenCut + Bilateral Solver results is shown in (c). (d) is the ground truth.

Table 4: **Analysis of τ .** We report CorLoc for unsupervised single object discovery on VOC07, VOC12, COCO20K, and Top-1 Loc for weakly supervised object detection on CUB and ImageNet-1K.

τ	VOC07	CorLoc		Top-1 Loc	
		VOC12	COCO20K	CUB	ImageNet-1K
0	67.4	71.3	56.1	73.0	53.8
0.1	68.6	72.1	58.2	73.2	53.4
0.2	68.8	72.1	58.8	72.9	52.3
0.3	67.7	72.1	58.2	70.8	50.4

[Internet Images]

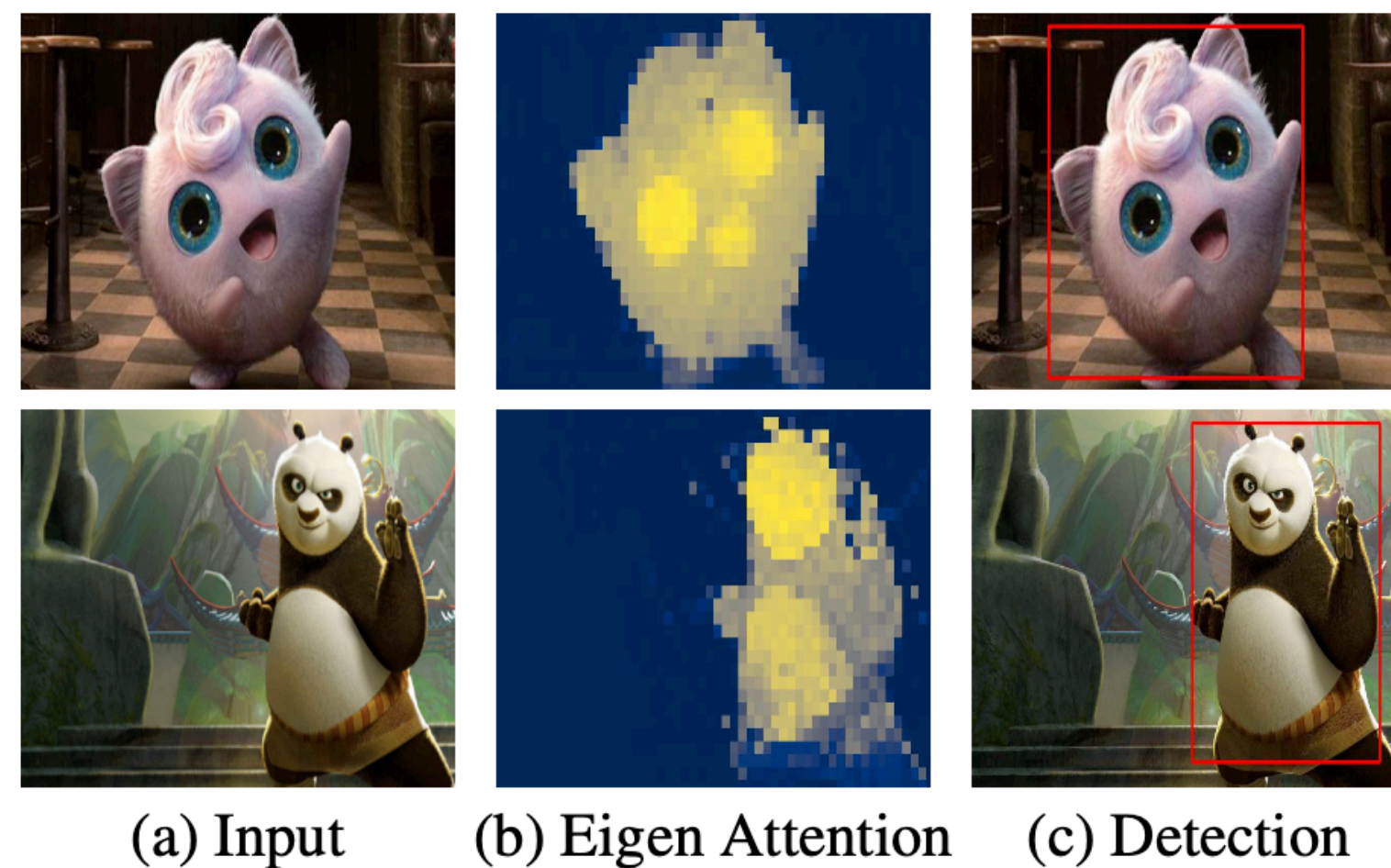


Figure 5: **Visual results of images taken from the Internet.** We show the input images, our eigen attention and final detection in (a) (b) (c) respectively.

[Limitations]

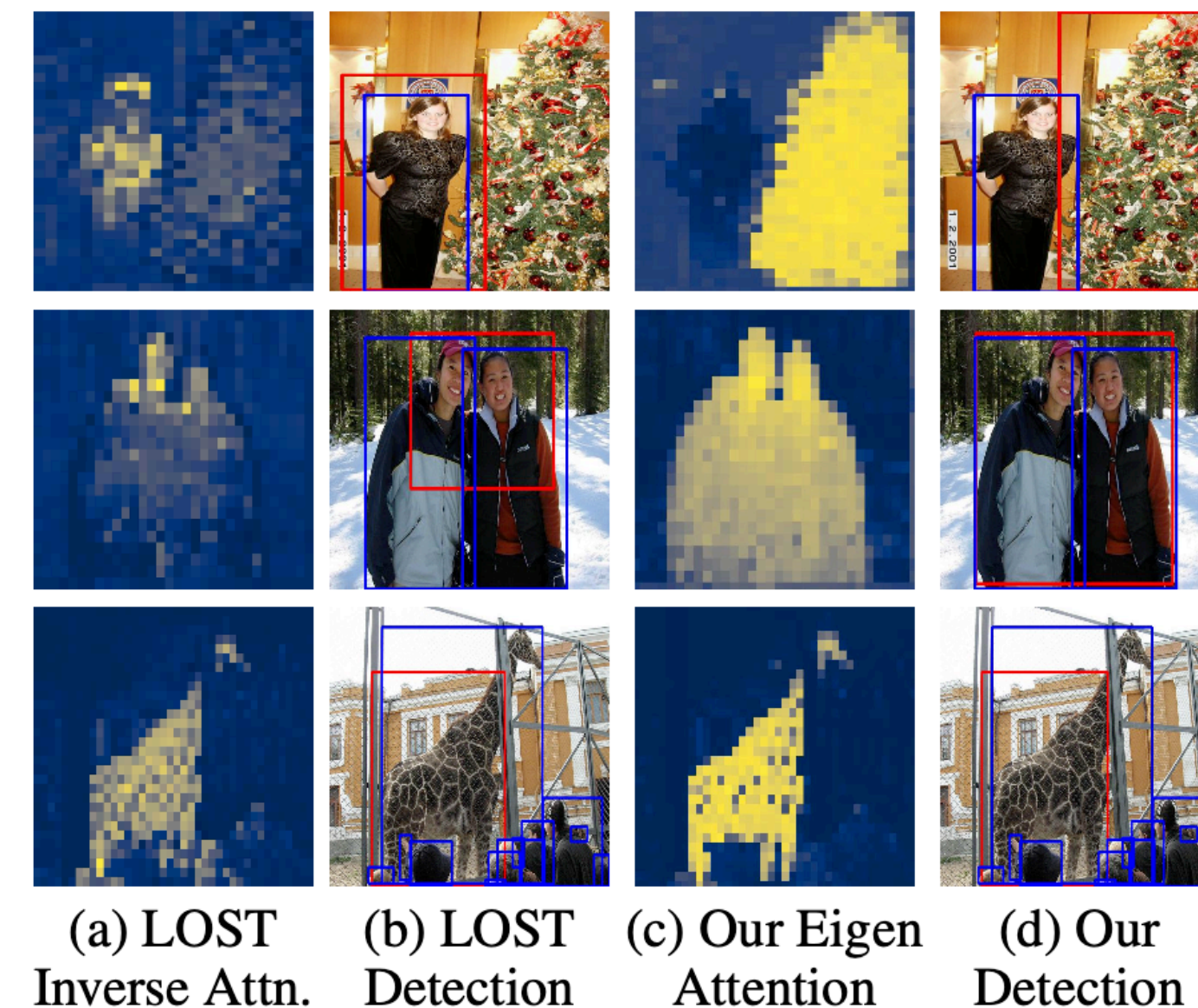
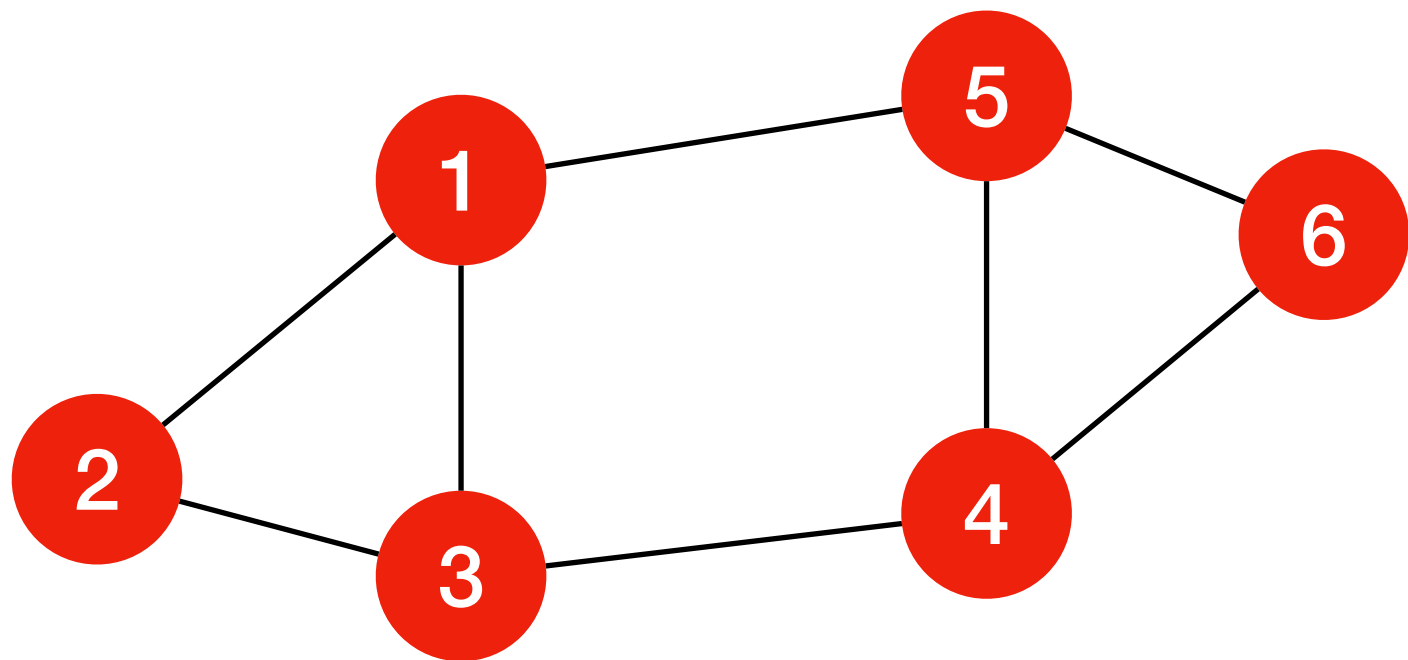


Figure 6: **Failure cases on VOC12 (1st and 2nd row) and COCO (3rd row).** LOST [45] mainly relies on the map of inverse degrees (a) to perform detection (b). For our approach, we illustrate the eigenvector in (c) and our detection in (d). **Blue** and **Red** bounding boxes indicate the ground-truth and the predicted bounding boxes respectively.

Laplacian Matrix

- <https://darkpgmr.tistory.com/105>
- <https://www.youtube.com/watch?v=Nvc7ZRVjciM&list=PL5yujGYFVt0BCu7DXfEgD7M51Tj6S7s4A&index=3>
- <https://junklee.tistory.com/112>
- https://www.youtube.com/watch?v=c0_vNfNZ4JM&list=PLLssT5z_DsK9JDLcT8T62VtzwyW9LNepV&index=28
- (28 ~ 34)
- https://en.wikipedia.org/wiki/Laplacian_matrix



	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	3	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

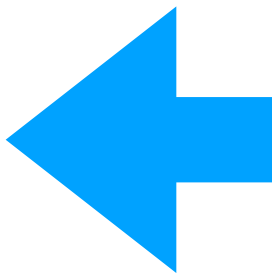
Degree matrix (D)

	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	1	1
5	1	0	0	1	0	1
6	0	0	0	1	1	0

Adjacency matrix (A)

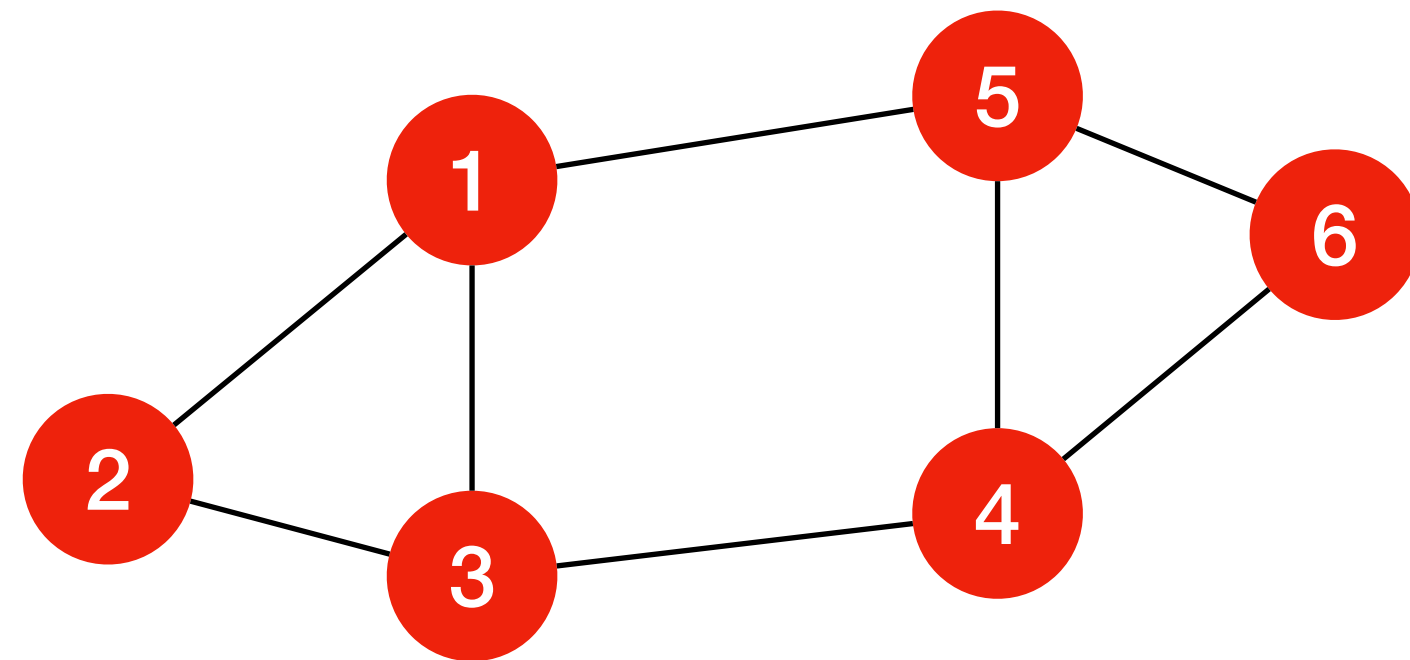
	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix (L)



$$L = D - A$$

$$L_{ij} = \begin{cases} deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$



	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	3	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

Degree matrix (D)

	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	1	1
5	1	0	0	1	0	1
6	0	0	0	1	1	0

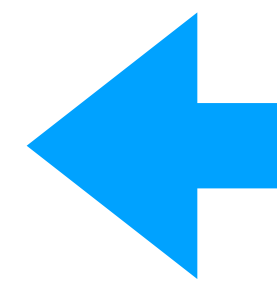
Adjacency matrix (A)

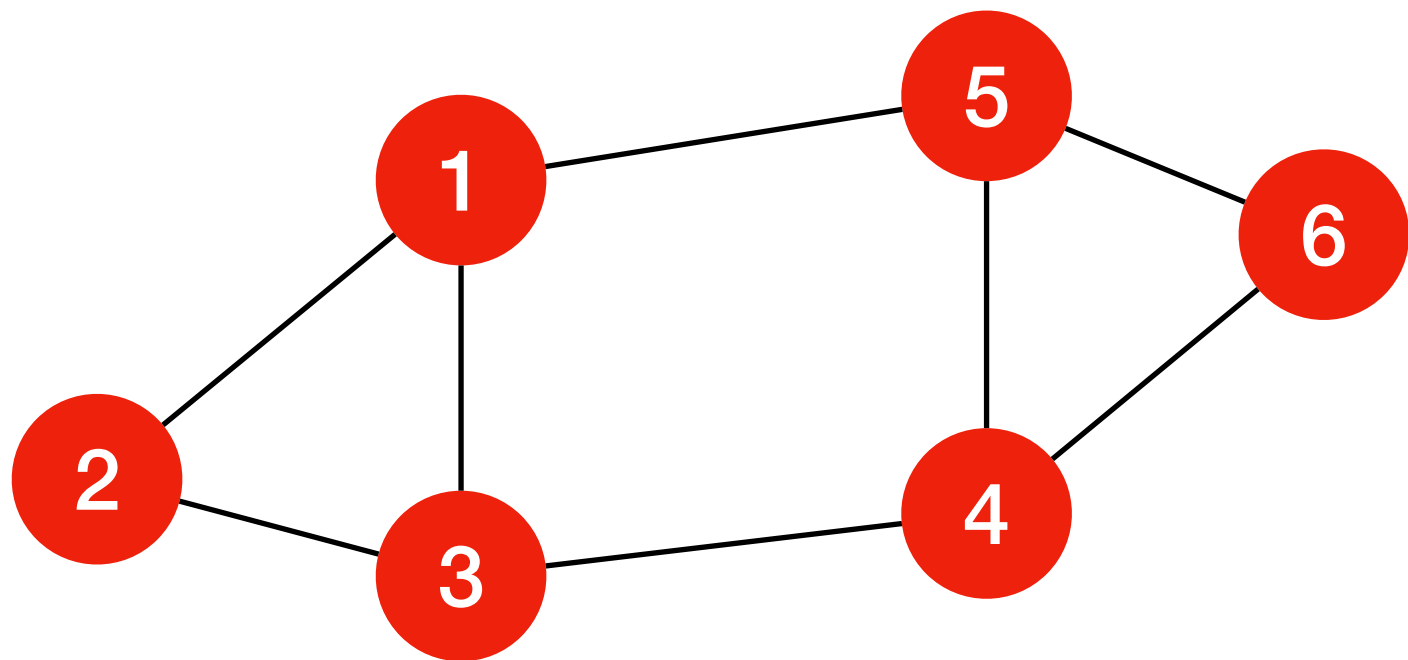
	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix (L)

$$Lx = \lambda x$$

- Symmetric Matrix
- Eigenvalues(λ) are non-negative real numbers
- Eigenvector(x) are real and orthogonal





$$Lx = \lambda x$$

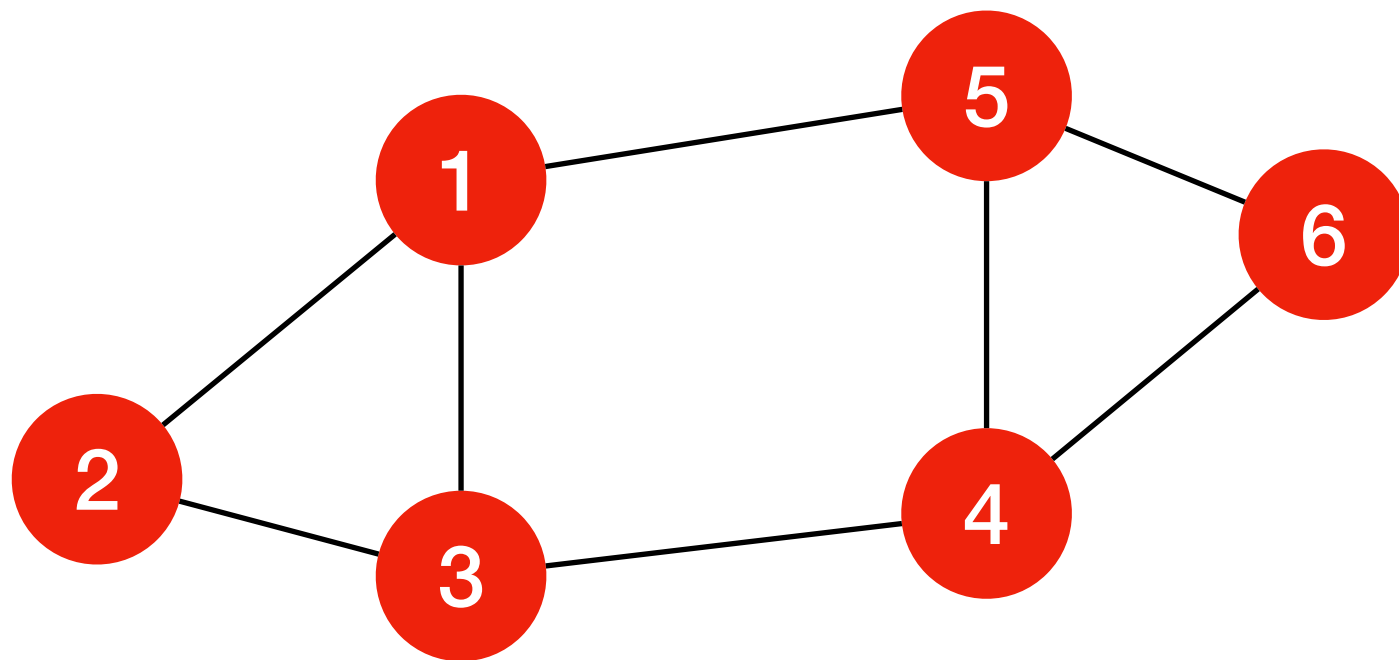
	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix (L)

smallest eigenvalue

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_1 = 0$$



	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

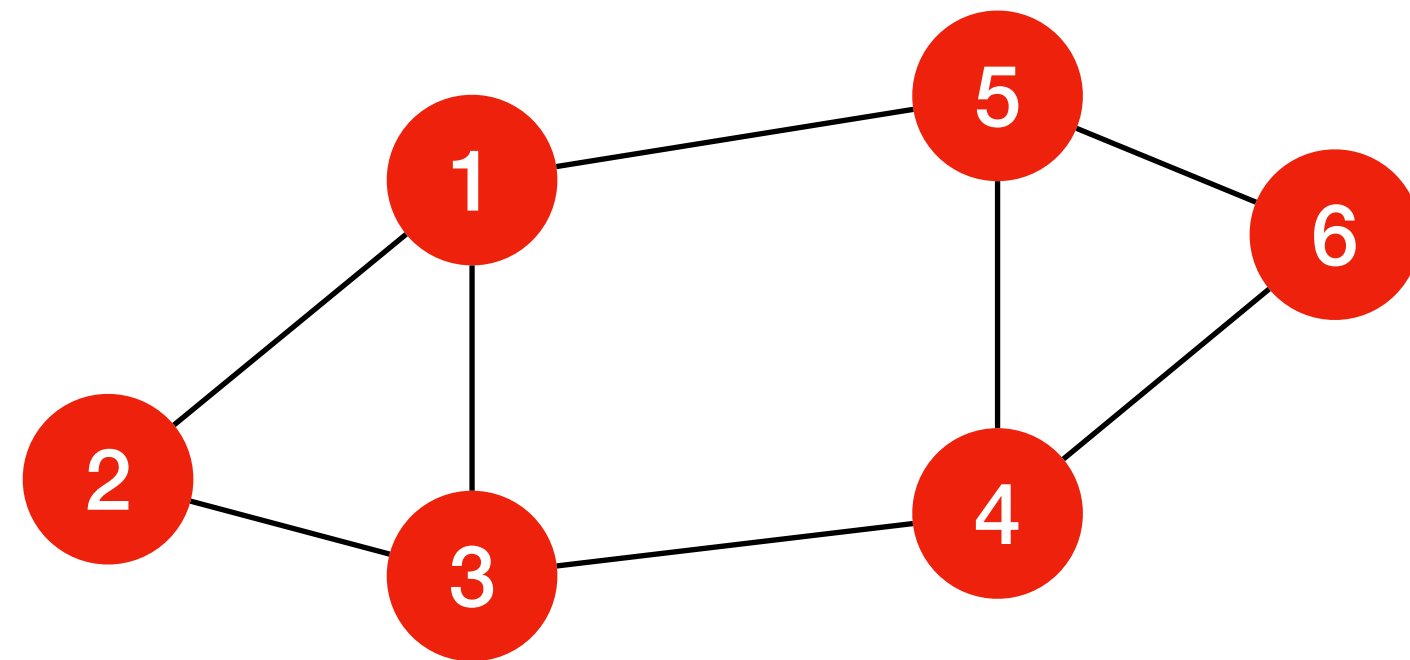
Laplacian matrix (L)

For symmetric matrix L :

$$\lambda_2 = \min_x \frac{x^T L x}{x^T x}$$

$$= \min_x \frac{\sum_{(i,j) \in E} (x_i - x_j)^2}{\sum_i x_i^2}$$

- $x^T L x = \sum_{i,j=1}^n L_{ij} x_i x_j = \sum_{i,j=1}^n (D_{ij} - A_{ij}) x_i x_j$
- $= \sum_i D_{ii} x_i^2 - \sum_{(i,j) \in E} 2x_i x_j$
- $= \sum_{(i,j) \in E} (x_i^2 + x_j^2 - 2x_i x_j) = \sum_{(i,j) \in E} (x_i - x_j)^2$



	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix (L)

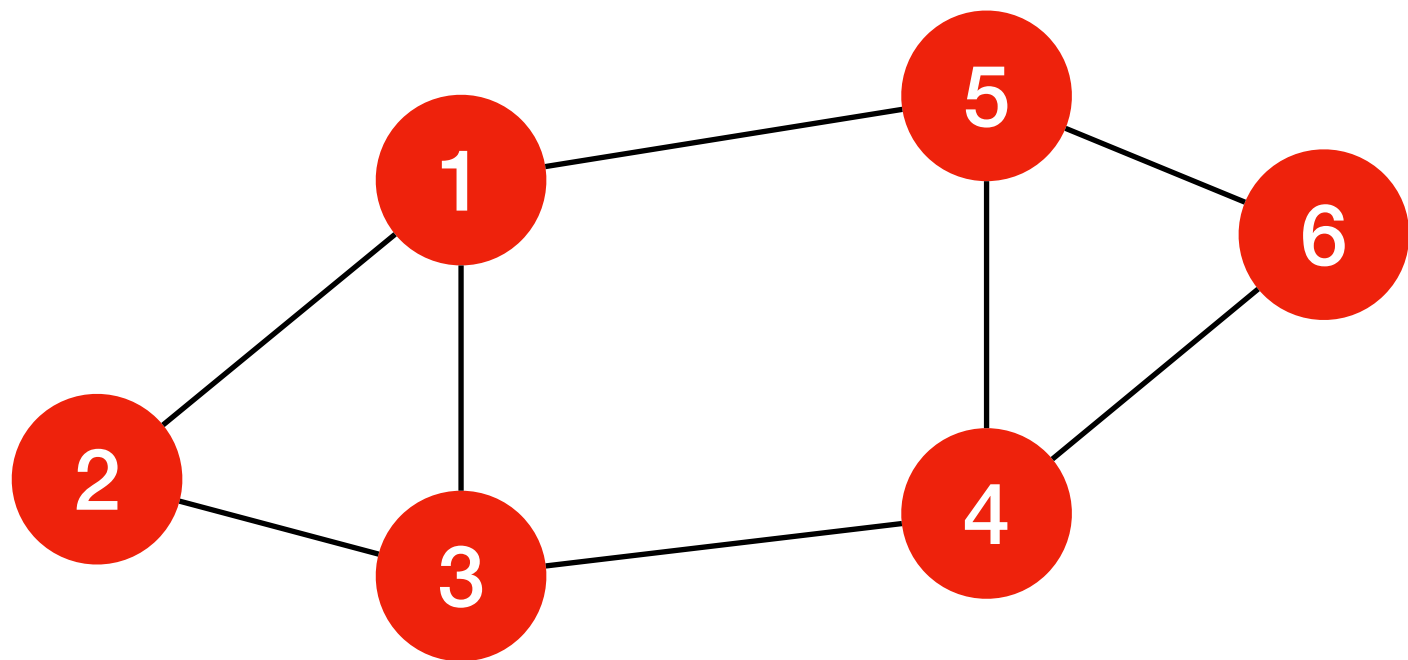
For symmetric matrix L :

$$\lambda_2 = \min_x \frac{x^T L x}{x^T x}$$

$$= \min_x \frac{\sum_{(i,j) \in E} (x_i - x_j)^2}{\sum_i x_i^2}$$

$$= \min_x \sum_{(i,j) \in E} (x_i - x_j)^2$$

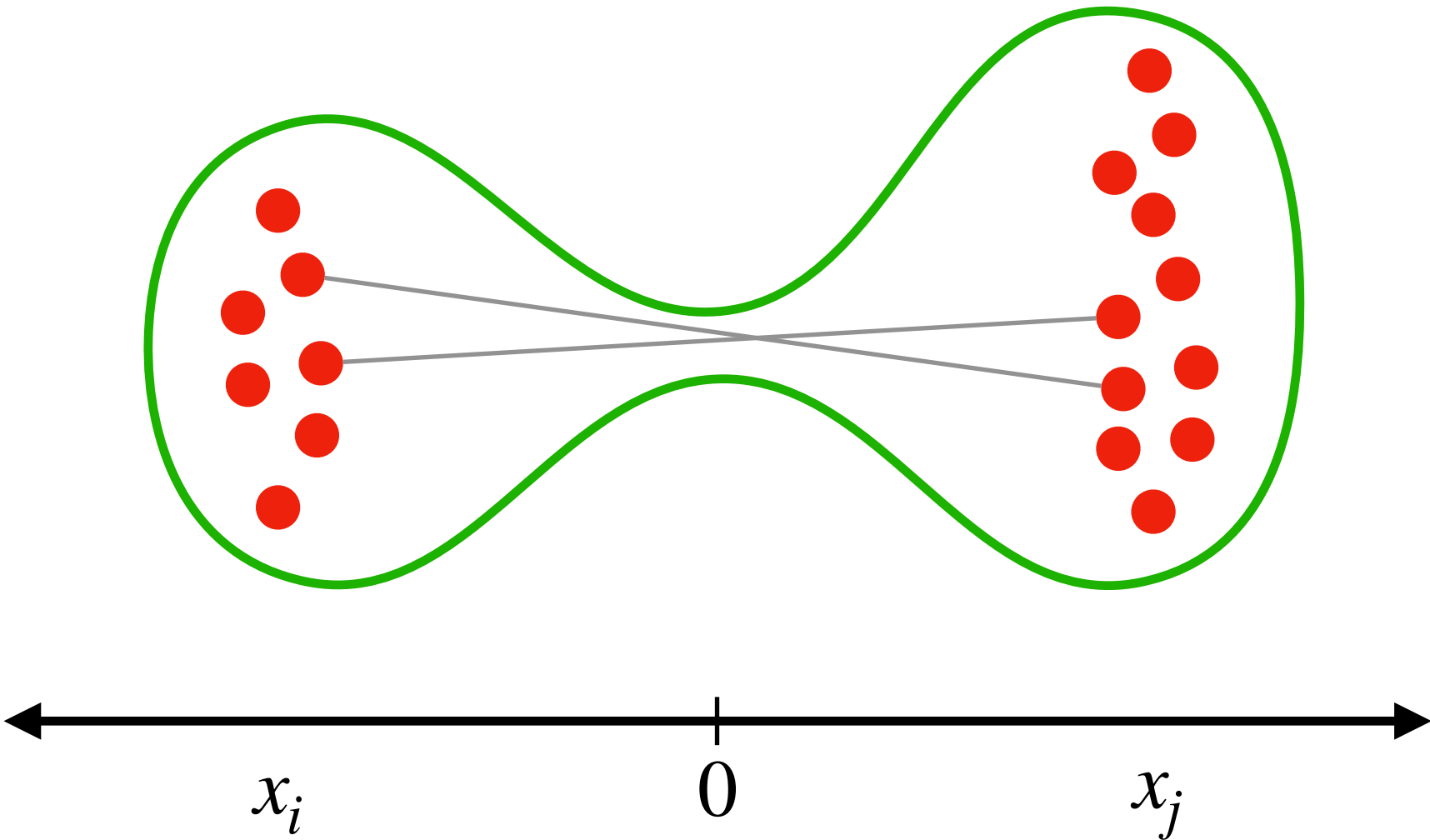
- x is unit vector: $\sum_i x_i^2 = 1$
- x is orthogonal to 1st eigenvector $(1, \dots, 1)$ thus $\sum_i x_i \cdot 1 = \sum_i x_i = 0$



	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix (L)

$$\lambda_2 = \min_x \sum_{(i,j) \in E} (x_i - x_j)^2$$



Thanks